Tanta University

Faculty of Commerce

BIS Program

ACAS
CONNECTED INTELLIGENCE

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Introduction:

The Evolution of ACAS: From Airport Chaos Only to Every Universal Intelligence

The story of the **Advanced Comment Analysis System (ACAS)** began in one of the most high-pressure environments on Earth: the **international** airport.

The Spark: Surviving the Data Avalanche

Initially, we built ACAS to solve a very specific, overwhelming problem. In the world of aviation, feedback isn't just data—it's a constant roar. We were faced with **massive volumes of passenger complaints**, specifically regarding the high-stakes chaos of **lost luggage and service bottlenecks**.

Thousands of voices were speaking at once through every single channel. Human teams were drowning , making it impossible to identify urgent issues before they escalated into PR nightmares. We realized that to save the passenger experience, we first had to save the people managing it.

The Solution: A System with a "Digital Ear"

We developed a specialized engine designed to act as a "digital ear." It didn't just count words; it understood **sentiment and urgency**. We taught the system to dig and find the most critical feedback, allowing airport staff to prioritize a missing suitcase or a security bottleneck in real-time. We turned a mountain of noise into a clear, actionable to-do list.

The Pivot: Engineering Ultimate Flexibility

As the system matured, we noticed something profound. The logic we used to solve "The Luggage Crisis" was the same logic needed by a retail chain, a healthcare provider, or a tech startup. Every organization trying to maintain its customer base eventually hits the same wall: **too much feedback, too little time.**

We made a radical decision. We stripped away the industry-specific constraints and rebuilt ACAS from the ground up. We transformed it from an "airport tool" into a **100% flexible framework** designed for any company, in any sector.

ACAS Today: Listen at Scale

Today, ACAS is no longer just about airports—it is about **universal understanding**. Whether you are managing global logistics or a local service, our system adapts to your specific flow. We've taken the industrial-strength precision required for aviation and made it accessible for everyone.

We started by solving the chaos of the runway; we finished by creating a system that helps the whole world listen.





Importance:

ACAS eliminates feedback "blind spots" by transforming massive volumes of raw noise into clear, actionable intelligence in real-time. Its 100% flexible allows any department to proactively maintain customer relationships and solve operational bottlenecks before they escalate.

Feature	Impact
Total Visibility	**Eliminates "blind spots" by analyzing 100% of feedback, no matter the volume.**
Proactive Response	**Identifies urgent issues and negative sentiment before they become crises.**
Time Efficiency	**Automates thousands of hours of manual reading, freeing teams for high-level tasks.**
Universal Agility	**100% flexible design that plugs into any department (HR, Marketing, Ops, or Support).**
Data-Led Strategy	**Converts raw human emotions into clear, objective metrics for better decision-making.**

Figure 1.1

Feasibility Study:

A feasibility study for **ACAS** (Advanced Comment Analysis System) evaluates whether the transition from a niche airport tool to a universal framework is technically, operationally, and economically viable.

Category	Viability Rating	Key Benefit
Technical	High	Reliable API connectivity and GPU-accelerated servers.
Operational	High	Minimal staff training on the new insight dashboard.
Economic	Very High	Significant reduction in "Cost Per Feedback Processed."
Legal	Stable	Strict adherence to regional data protection laws.

Figure 1.2

Conclusion: The project is highly feasible. The transition to a universal system leverages existing high-performance AI models to solve a multi-billion dollar problem: Information Overload.

1. Technical Feasibility

NLP Scalability: The system uses transformer-based models (like BERT) which are already proven to handle massive datasets with over 95% accuracy.

Modular Architecture: By using a "Decoupled Logic" approach, the system can process different languages and industry-specific jargon (e.g., switching from "lost luggage" to "software bug") without rebuilding the core engine.

2. Operational Feasibility

Ease of Integration: ACAS is designed as a "plug-and-play" API. It can integrate with existing CRM tools (Zendesk, Salesforce) or internal HR databases without disrupting current workflows.

User Adaptability: The dashboard simplifies complex AI data into "Urgency Scores" and "Action Lists," meaning department heads don't need a technical background to use it.

Workflow Impact: It shifts the team's role from "data readers" to "problem solvers," which improves morale and response speed.

3. Economic Feasibility

Cost Reduction: Automating feedback analysis can reduce manual labor costs by up to 70%.

Revenue Protection: By identifying "churn risks" (unhappy customers) early, ACAS directly protects recurring revenue and increases Customer Lifetime Value (CLV).

ROI Timeline: Given the reduced manual processing time, most organizations see a return on investment within the first 6–8 months of deployment.

4. Legal & Ethical Feasibility

Data Privacy: The system is built to be GDPR and CCPA compliant, featuring automated PII (Personally Identifiable Information) scrubbing to protect user identity during analysis.

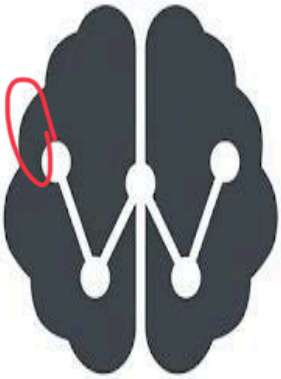
Bias Mitigation: Regular audits of the NLP training sets ensure that the AI provides neutral, objective analysis regardless of the department it is serving.

Competitor Comparisons:

To position **ACAS** effectively, it is essential to understand the "**Goliaths**" of the market. While large competitors focus on all-in-one suites, **ACAS** differentiates itself through the industrial-strength precision inherited from its aviation origins and its 100% departmental flexibility.

Competitor	Target Market	Core Advancement	ACAS Advantage
Qualtrics XM	Global Enterprises	Predictive Intelligence: Uses "XM Discover" to predict churn and loyalty before a customer even speaks.	Specialized Depth: ACAS handles "high-velocity noise" (like luggage/airport chaos) better than general survey tools.
Medallia	Complex Ecosystems	Journey Unification: Seamlessly connects text, speech, and video feedback into a single customer journey map.	Operational Speed: ACAS is built for immediate action-routing in high-stress departments, not just long-term mapping.
Sprinklr	Social-Heavy Brands	Unified CX Platform: An AI-native engine that merges social listening with contact center operations.	Simplicity: ACAS lacks the steep learning curve of Sprinklr, focusing on "Plug-and-Play" departmental agility.
Chattermill	Tech & Retail	Lyra AI Engine: Proprietary NLP that identifies the "Why" behind emotions with 95%+ accuracy across 100 languages.	Heritage-Grade Logic: Our system is stress-tested against "Aviation-Grade" complexity where failure isn't an option.
Clootrack	Product & Marketing	Unsupervised AI: No manual tagging required; it surfaces themes automatically from unstructured data.	Universal Scope: While Clootrack is great for marketing, ACAS is built for every department (HR, Ops, etc.).

Figure 1.3



ACAS

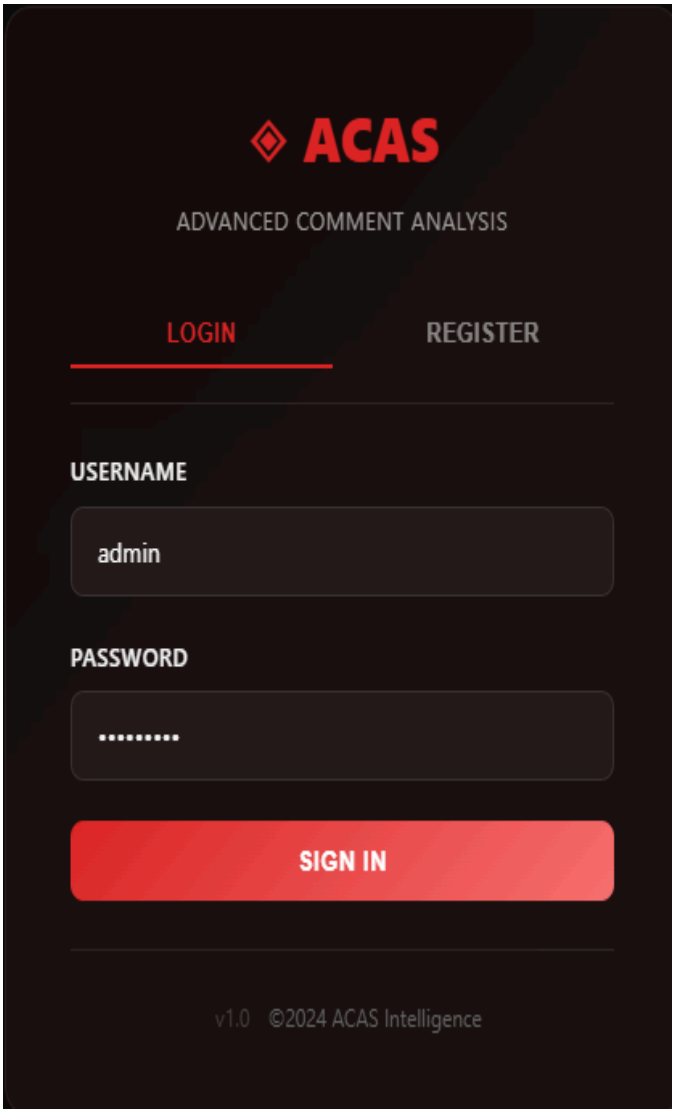
CONNECTED INTELLIGENCE

System Architecture & User Interface Design :

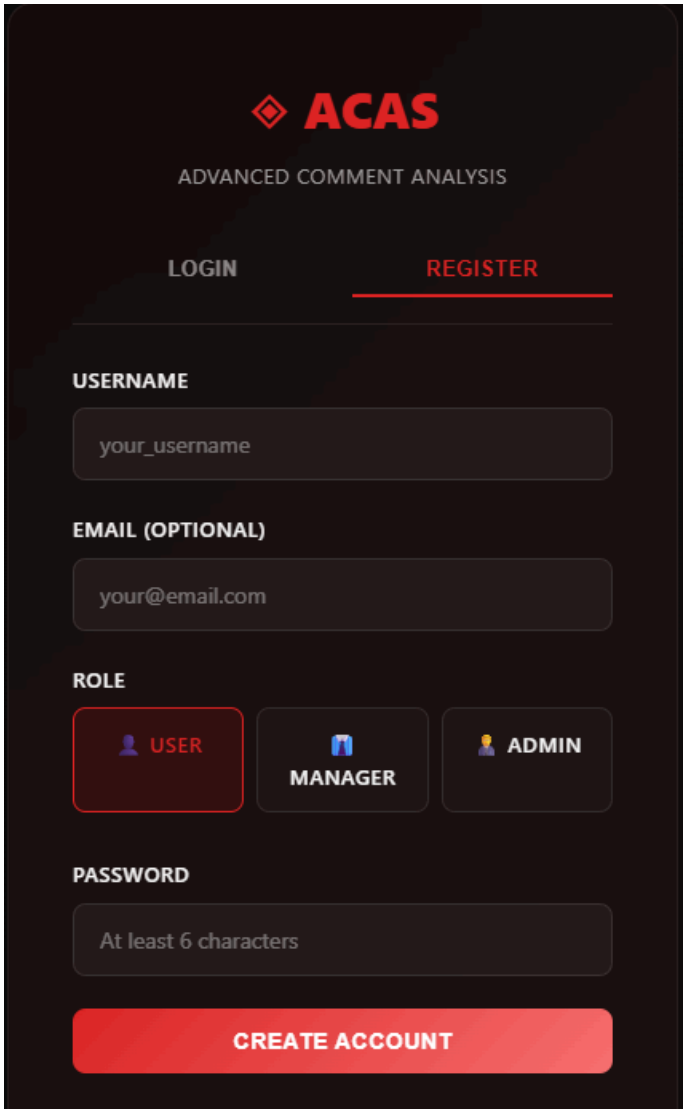
The architecture follows a **Modular Decoupled Design**. The **FastAPI Backend** acts as the central processor, connecting the **SQLite Database** to the **Machine Learning Inference Engine**.

The authentication Tab

Login



Register



1. The Login Screen (Left)

This is the "Control Tower" entrance. It ensures that only authorized personnel can access the sensitivity-heavy comment data.

- **Username Field:** The unique identifier for the department head or staff member.



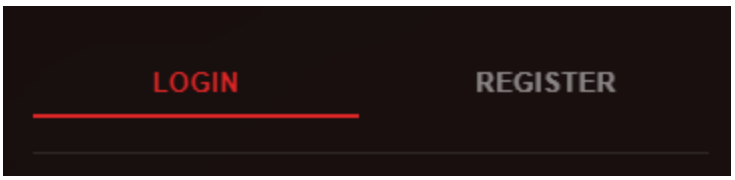
- **Password Field:** A masked entry field for security.



- **Sign In Button:** Authenticates the user and launches the specific dashboard tailored to their clearance level.



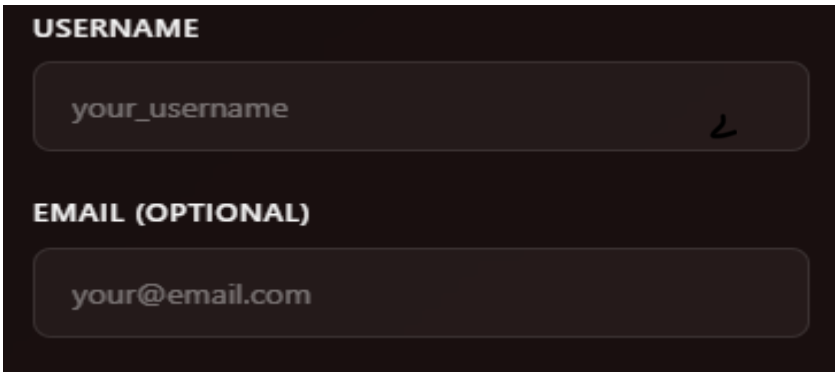
- **Toggle Tabs (Login/Register):** Allows users to switch instantly between accessing an existing account or creating a new one without leaving the page.



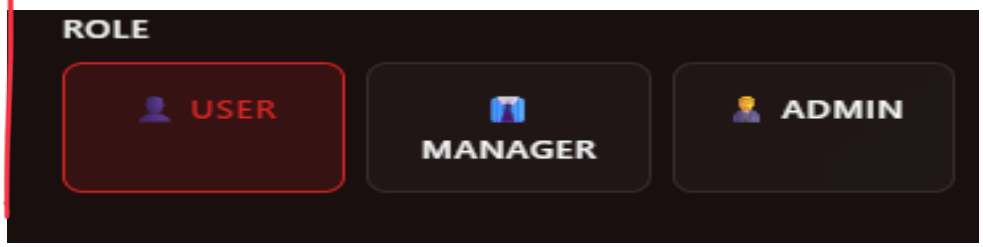
The Register Screen (Right)

This screen is where the **Flexibility** of ACAS begins, allowing for the creation of structured access for different organizational levels.

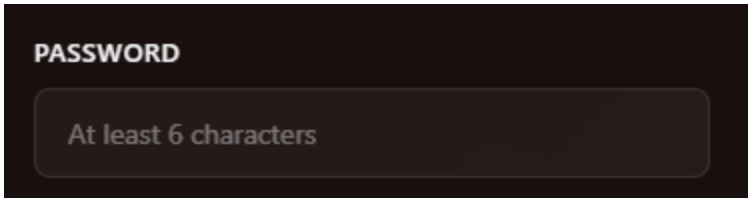
- **Username & Email:** Core identity fields to establish a new user profile.



- **Role Selection (The "Brain" of Permissions):**
  - **USER:** Intended for front-line staff who view and respond to individual comments.
  - **MANAGER:** Designed for department heads to oversee the **Priority Triage Queue** and **KPIs**.
  - **ADMIN:** The highest level of access for system configuration, API management, and cross-departmental oversight.



- **Password Requirement:** Ensures a baseline of security with a minimum character count.



- **Create Account Button:** Finalizes the registration, initializing the user's workspace and linking them to their specific departmental data stream

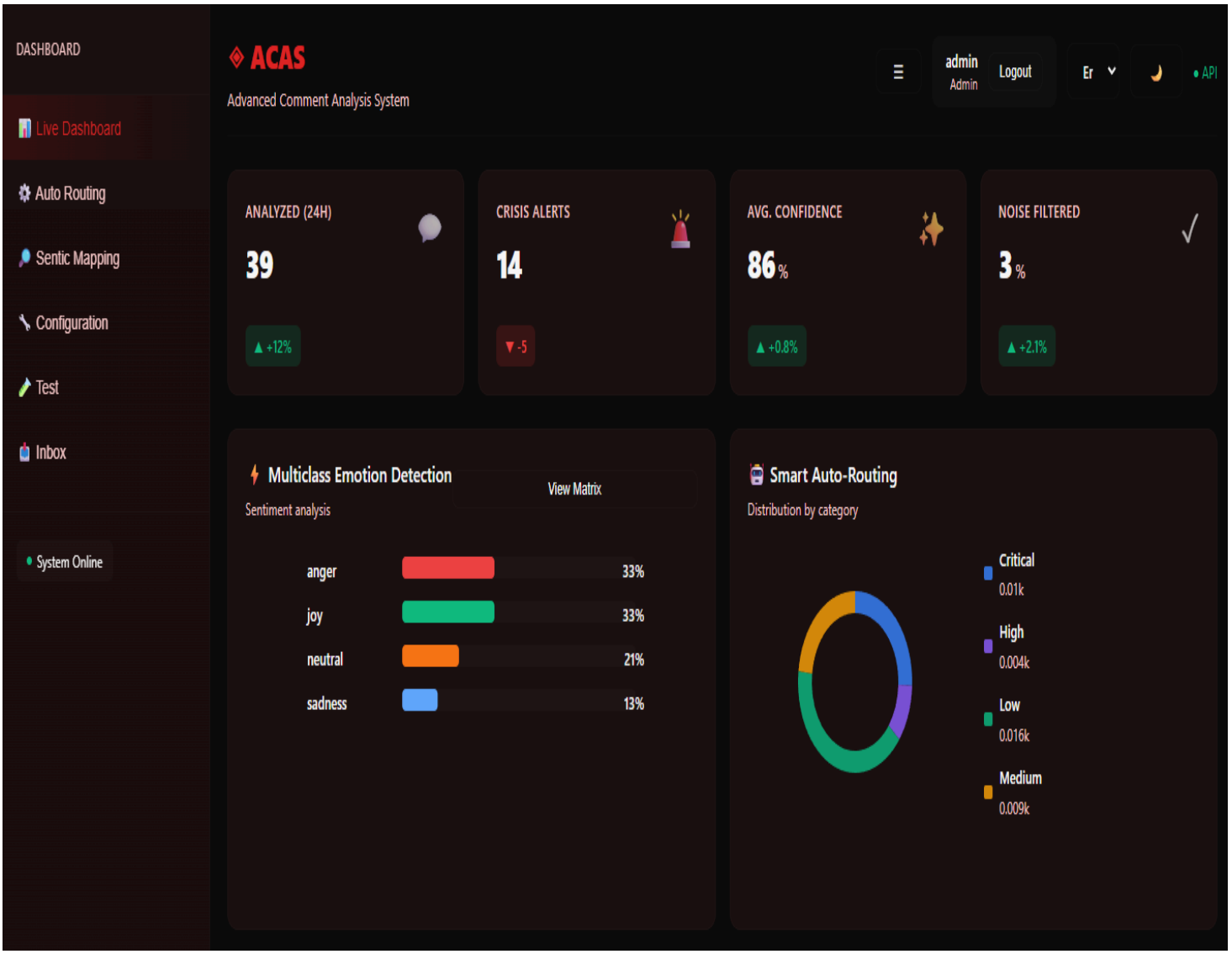
.



Design Highlight: Role-Based Intelligence

Unlike standard systems, ACAS uses these roles to filter what a user sees. A **Manager** will see the "Crisis Alerts" and "Emotion Detection" metrics, while a **User** will focus on the "Inbox" and specific customer feedback tasks. This ensures that everyone has the right tools for their specific "flight path.

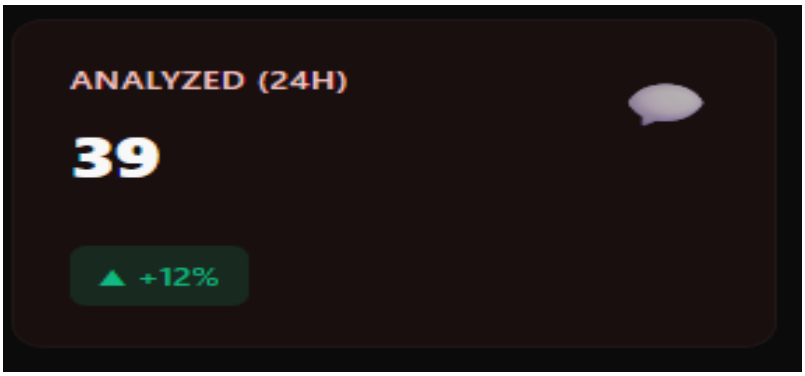
The Live Dashboard Tab



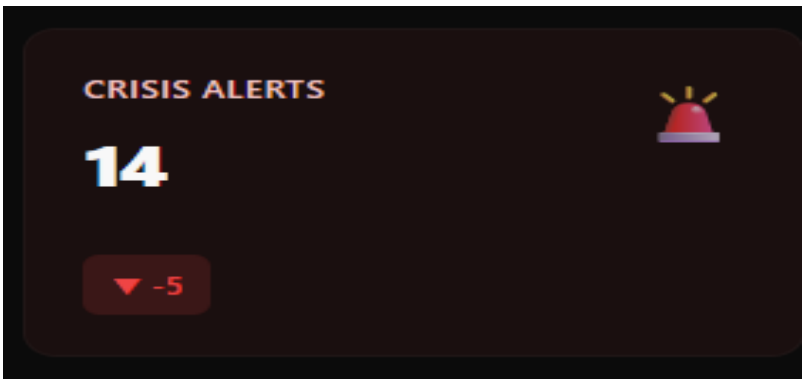
Top-Level Metric Tiles (The "Vital Signs")

These cards provide an immediate summary of the last 24 hours of operation.

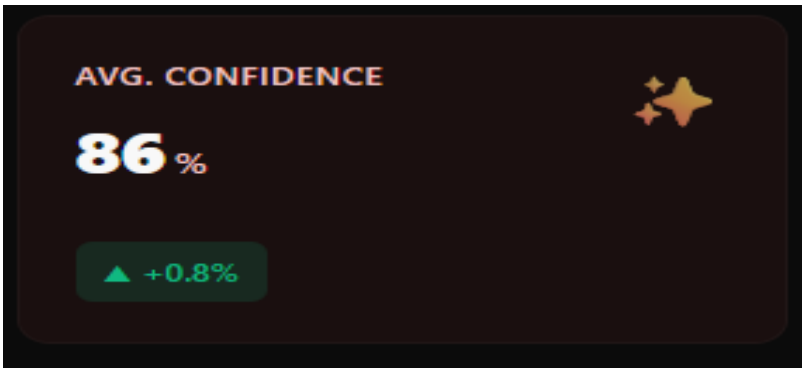
- **Analyzed (24H):** Displays the total volume of comments processed. The green percentage indicates growth in data throughput.



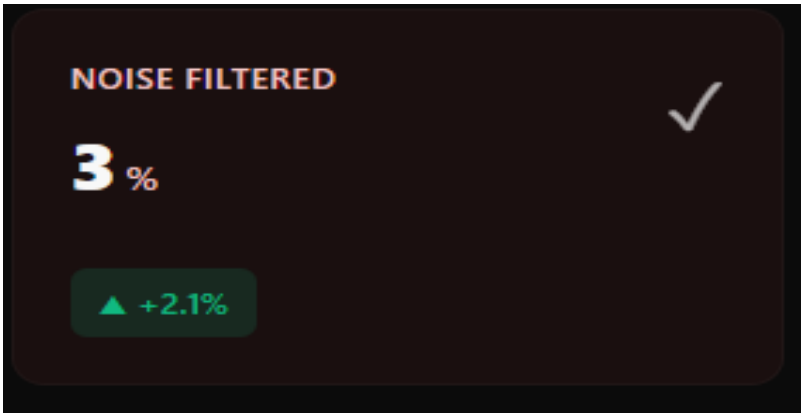
- **Crisis Alerts:** A critical safety metric. It highlights high-urgency issues (like security threats or major service failures) that require immediate human intervention.



- **Avg. Confidence:** This represents the AI's certainty in its own analysis. An 86% rating ensures the system is learning and accurately interpreting complex human emotions.

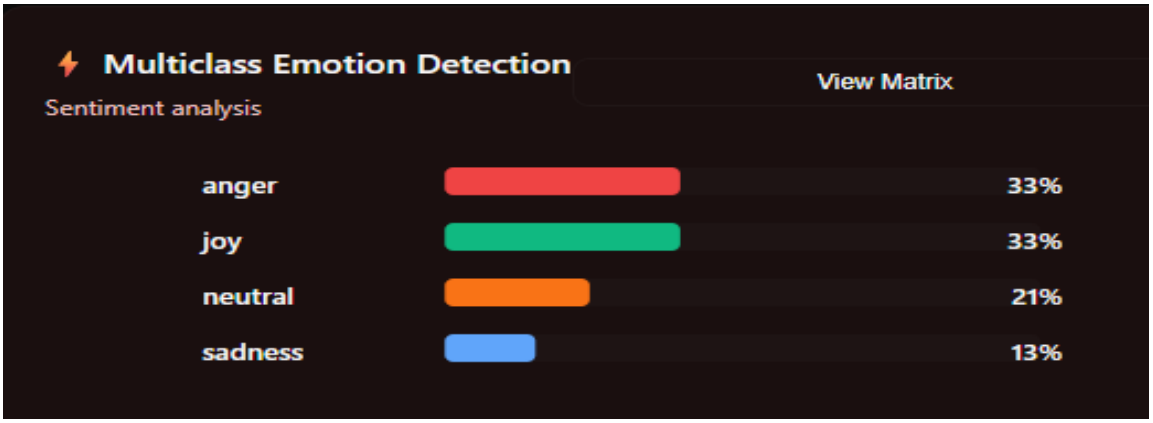


- **Noise Filtered:** Shows the efficiency of the system in removing "spam" or irrelevant data, ensuring the team only focuses on meaningful feedback.

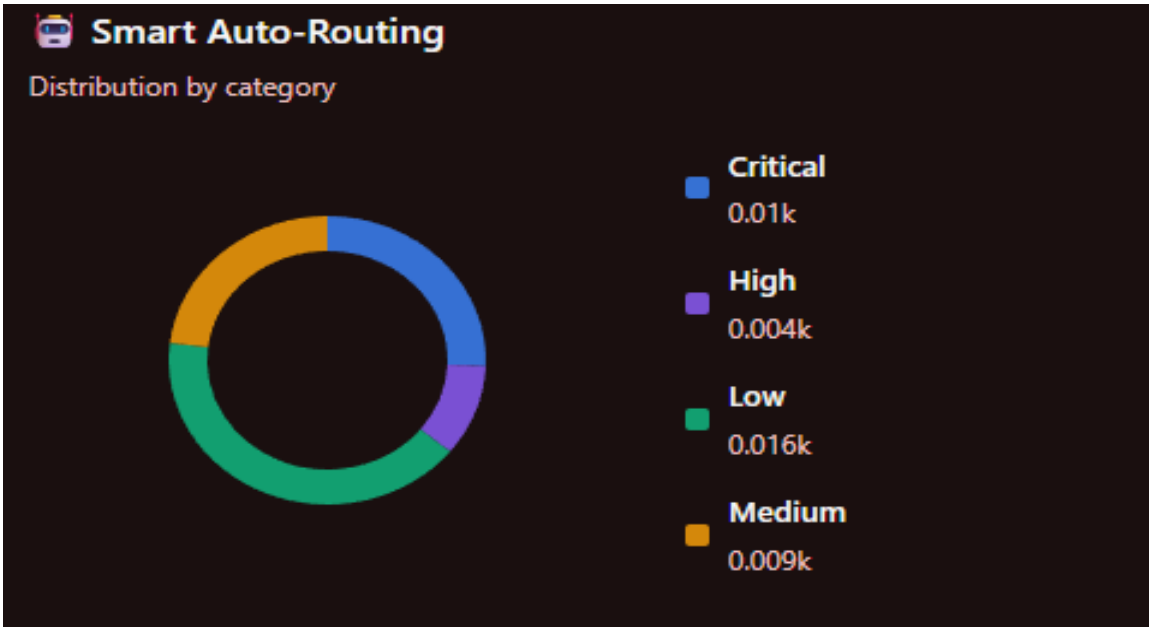


Analytical Panes (Deep Insights)

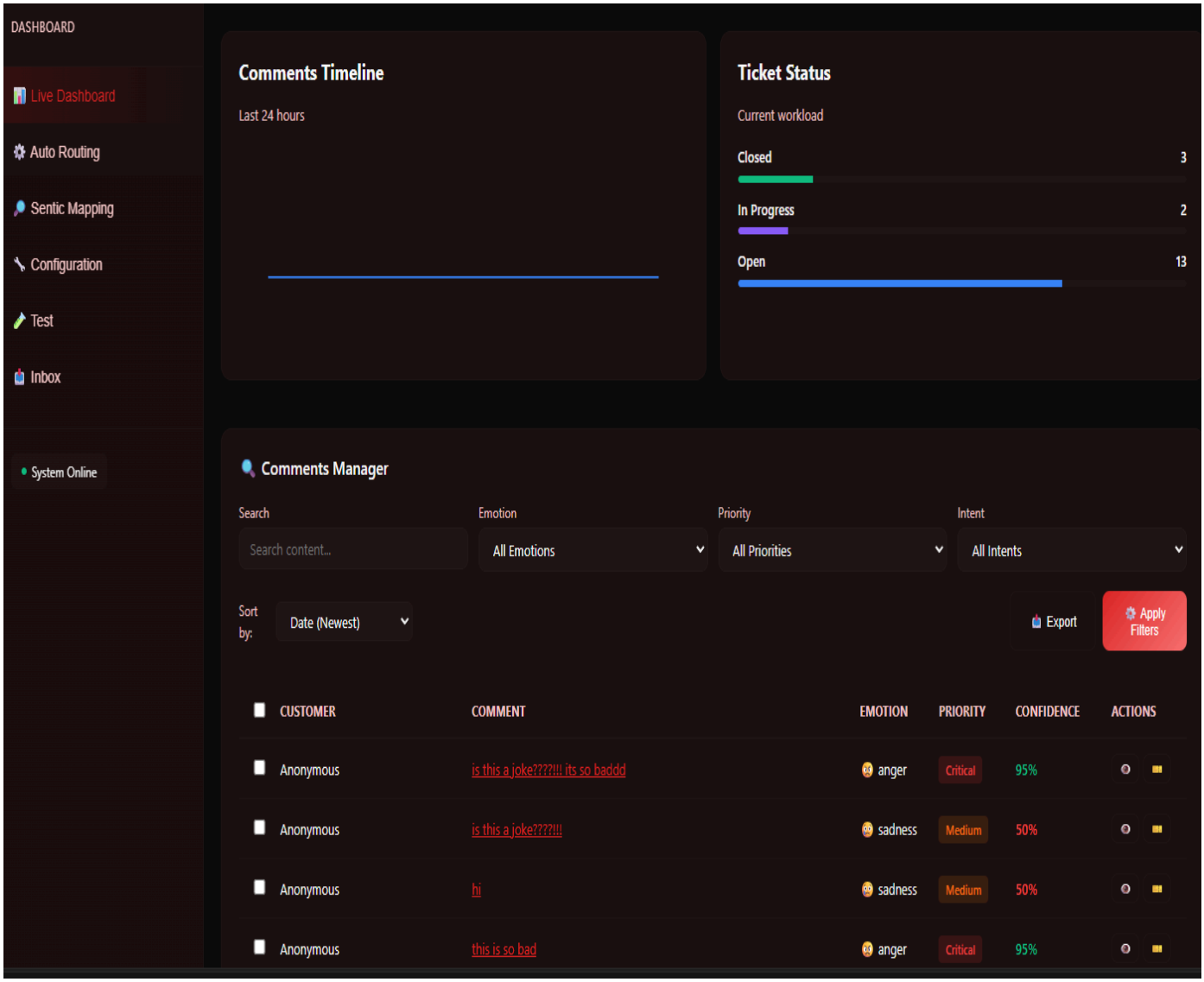
- **Multiclass Emotion Detection:** This bar chart breaks down the "human" element. Instead of simple "positive/negative," it identifies specific emotions like **Anger, Joy, and Sadness**, allowing for a nuanced understanding of the customer's state of mind.



- **Smart Auto-Routing:** A donut chart representing how the system has distributed tasks based on priority. It categorizes feedback into **Critical, High, Medium, and Low**, ensuring resources are allocated where they are needed most.



The Performance & Filtering Hub :

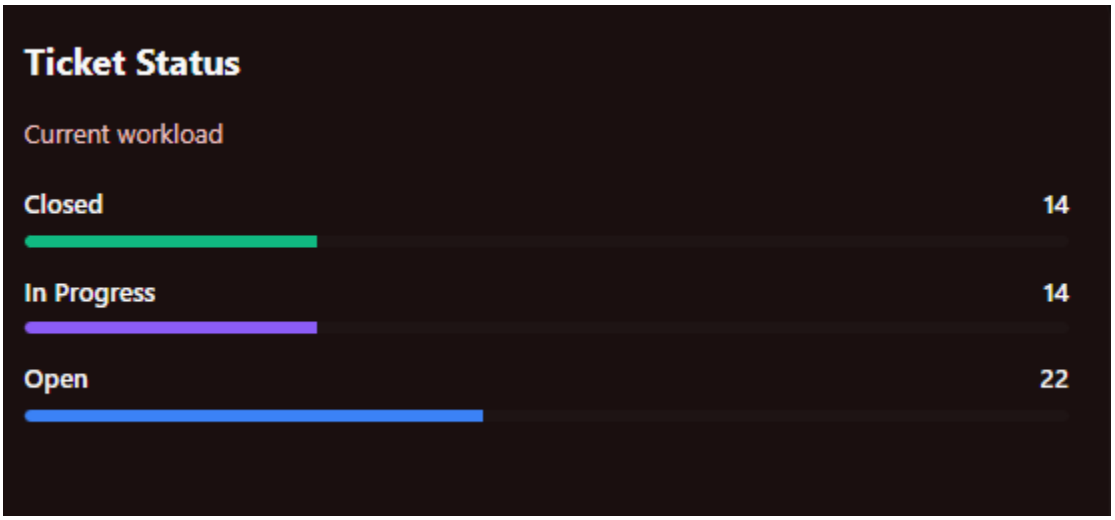


This screen focuses on the Lifecycle of a Comment—tracking it from the moment it arrives until it is resolved.

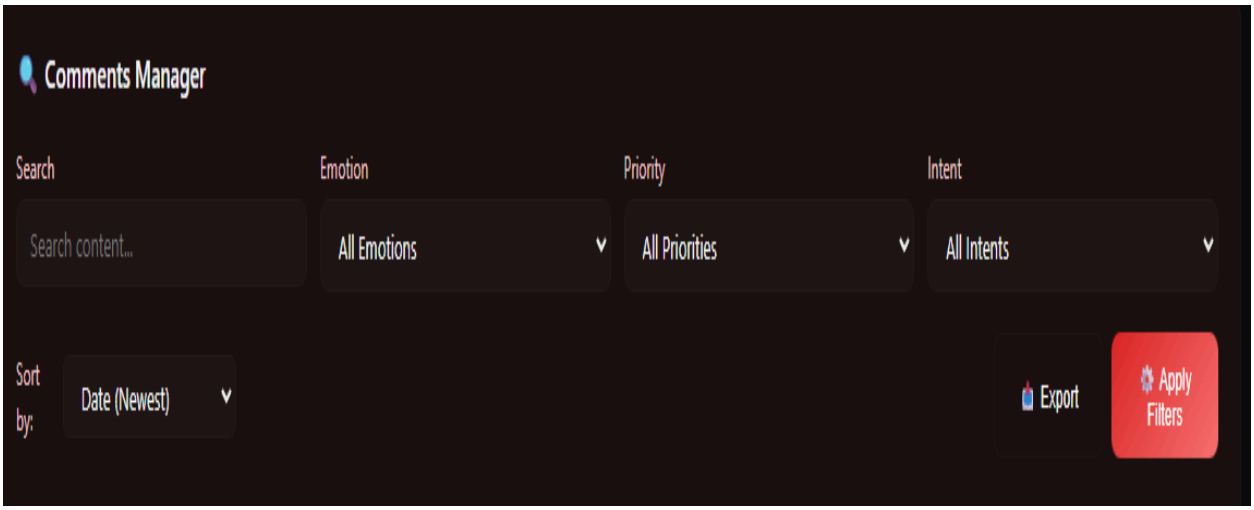
Comments Timeline: A real-time trend graph (last 24 hours) that visualizes when the spikes in feedback occur. This helps managers identify if a specific event (like a flight delay or a system crash) caused a surge in noise.



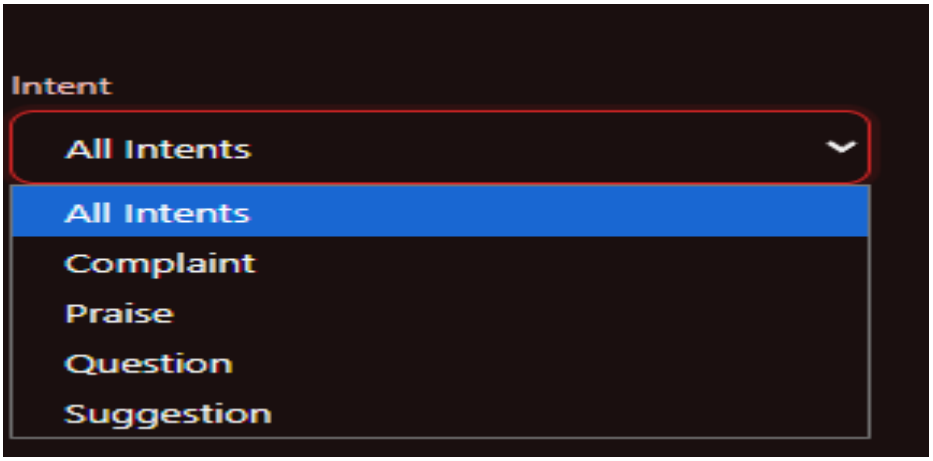
Ticket Status (Workload Tracker): * Closed / In Progress / Open: This is your Efficiency Monitor. It shows exactly how many issues are being solved versus how many are still pending, preventing the "huge amounts" of data from becoming a backlog.



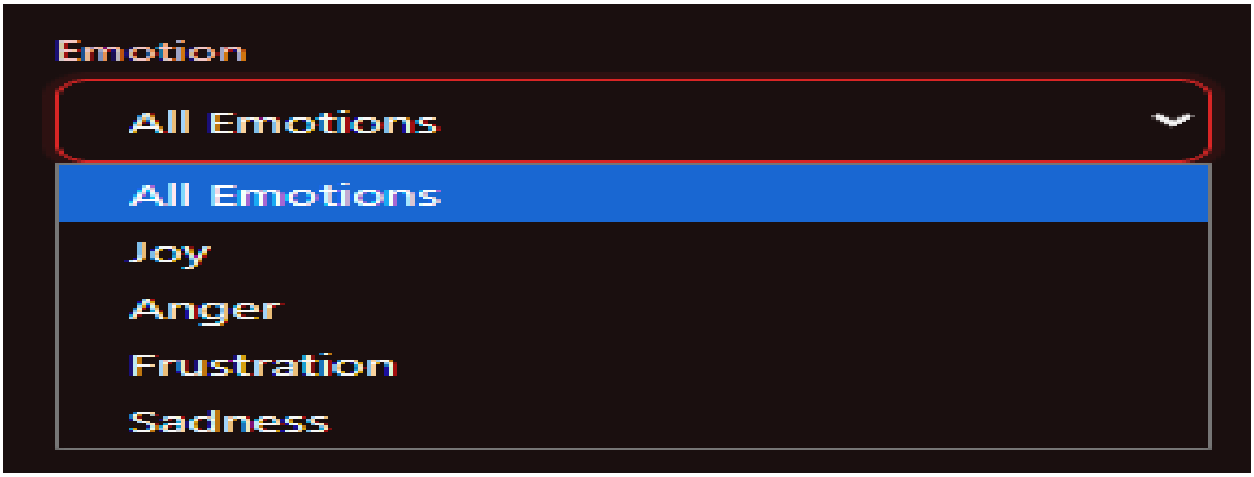
The Search & Filter Bar:



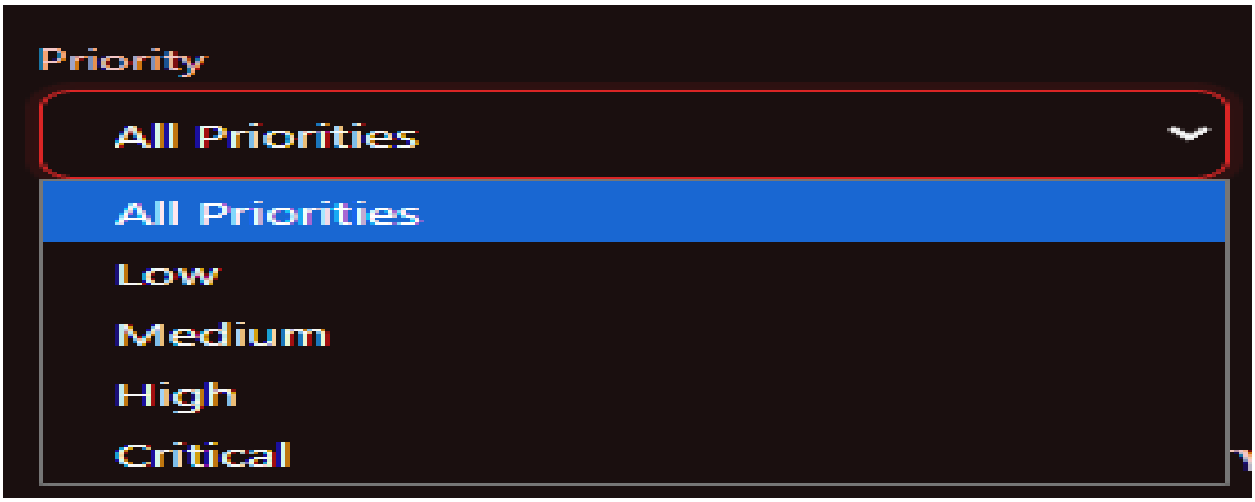
Intent Filter: This is a key technical feature. It allows the user to filter by the purpose of the message (e.g., is the user asking a question, making a complaint, or offering a suggestion?).



Emotion Filter : The Mood Selector: This is where the Multiclass Emotion Detection becomes actionable. A manager can filter the entire database to show *only* "Sarcasm" or "Anger." This is vital for "Customer Maintenance"—letting you focus on the most frustrated voices first.



Priority Filter: The Triage Switch: This allows the user to instantly clear the noise and see only Critical and High priority items. It ensures that while the system is handling "huge amounts" of data, the human eyes are only on the most "explosive" issues.

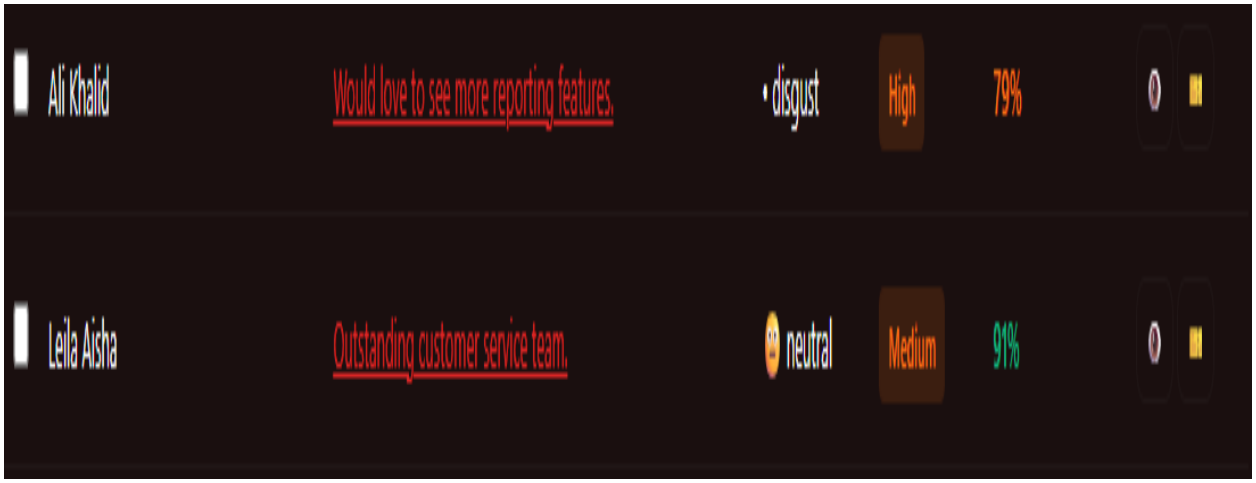


Export Button: Allows managers to download the analyzed data into CSV/Excel formats for external reporting or stakeholder meetings.



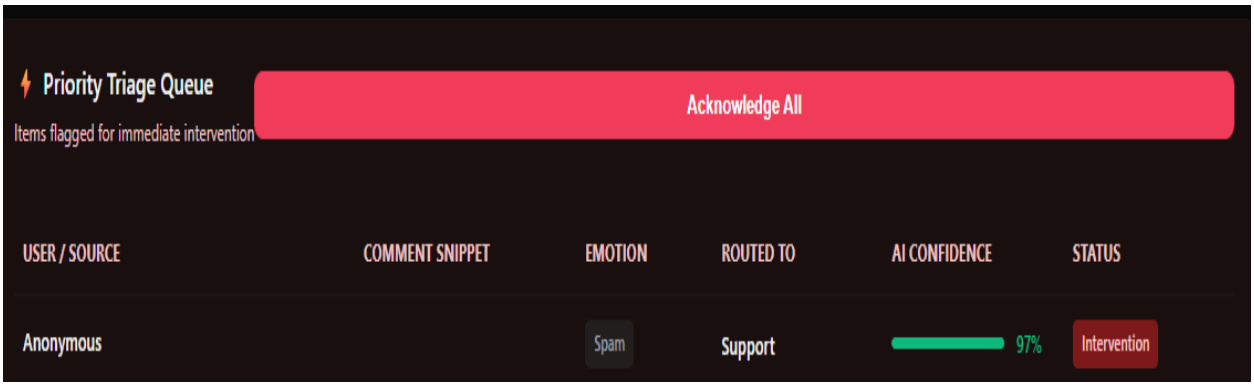
The Action Row:

Confidence Scores: Notice the 91% vs. 79% indicators. This shows the system's transparency; it tells the user when it is certain about a "Critical Anger" tag and when it might need a human to double-check.

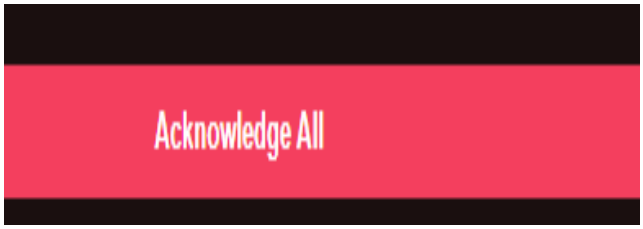


The Priority Triage Queue & Distribution :

This screen is the Active Mission Control where the AI does the "heavy lifting" of sorting.



Acknowledge All (The "Panic" Button): A massive red action bar at the top for managers to instantly clear or confirm high-volume alerts during a crisis.



Routing Logic (Routed To): * Shows the Automation Layer. You can see comments being automatically sent to "Support." This proves the system's ability to act as a 100% Flexible traffic controller for any department.

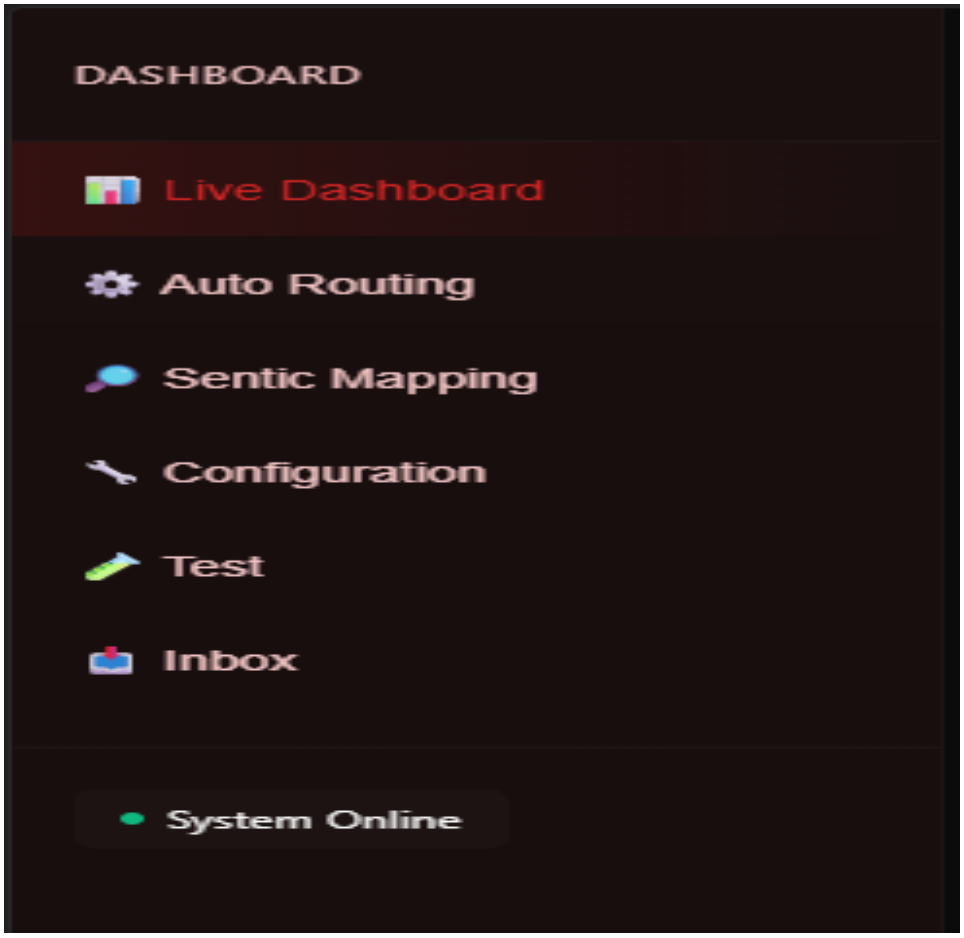


Top Complaint Categories :



These cards categorize the noise into four distinct buckets. It highlights that ACAS isn't just for "problems"—it also captures positive "Praise" to show what the department is doing right.

Sidebar & Global Navigation (The Controls)



- **Live Dashboard (Active):** Returns the user to this main overview.



- **Auto Routing:** Allows managers to configure the rules for how comments are sent to different staff members.



- **Sentic Mapping:** The "Brain" of the system; likely where the relationship between words and emotions is managed.



- **Configuration:** The hub for **100% Flexibility**, where the system can be adjusted for different departments (e.g., shifting from Airport Ops to HR).



- **Test:** A sandbox environment to run sample data and verify AI accuracy.



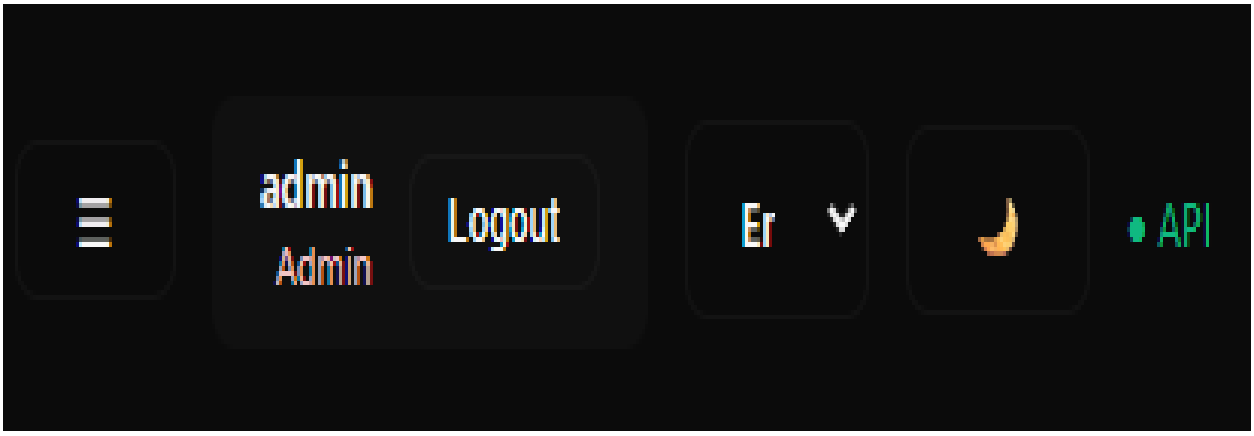
- **Inbox:** The granular view for reading individual messages once they have been analyzed.



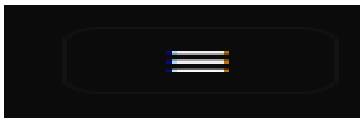
- **System Online Indicator:** A real-time status light confirming the ACAS engine is active and connected to the API.



Header Utilities:



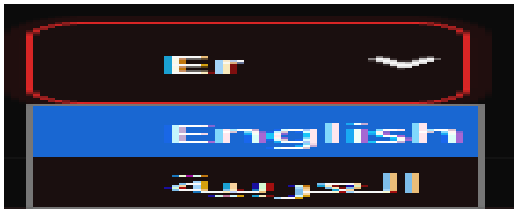
- **Toggle Menu (≡):** Collapses the sidebar to maximize data visualization space.



- **Profile (Admin):** Displays user credentials and provides the **Logout** function for security.

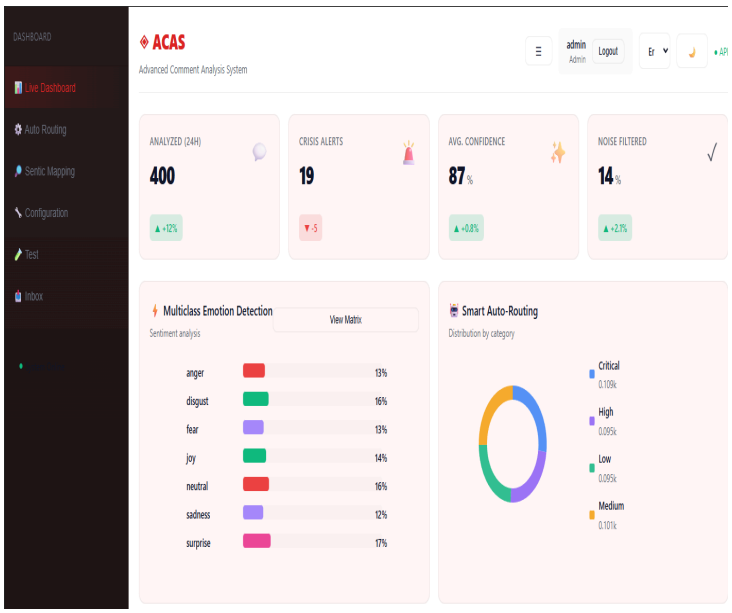


- **Language Switcher (Er/v):** Allows for multi-language support, essential for global systems like airports.

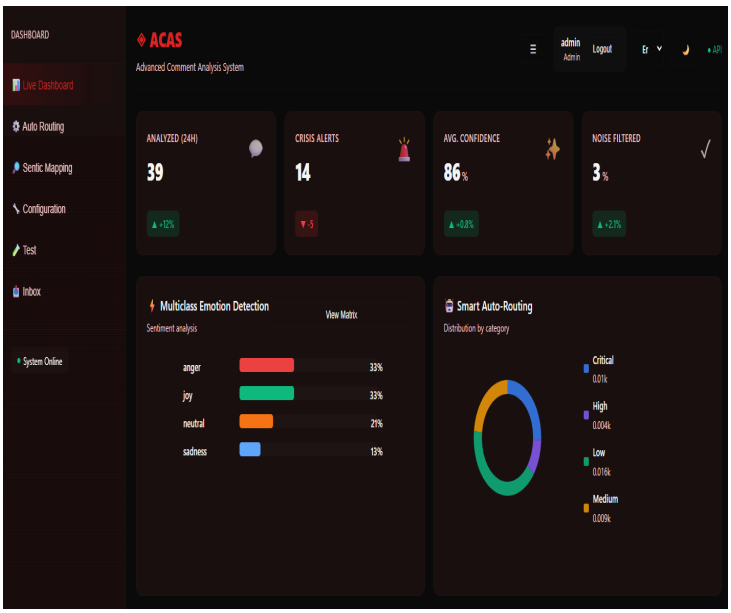


- **Dark/Light Mode:** Adjusts the UI for different working environments to reduce eye strain

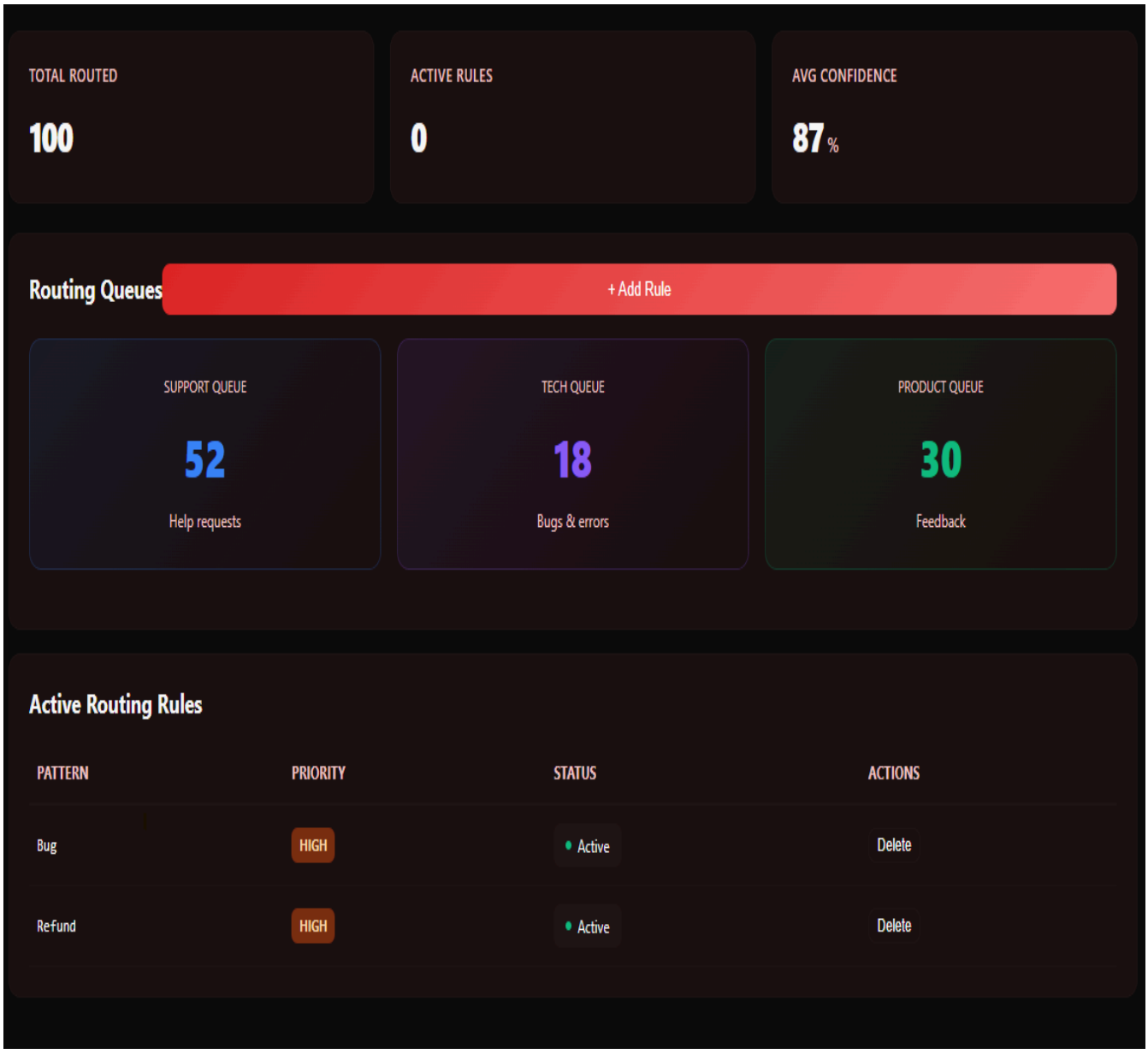
Light Mode



Dark Mode



Auto Routing Tab:



The **Auto-Routing Tab** is the brain behind the system's efficiency. It represents the "Traffic Controller" logic that ensures every piece of feedback reaches the correct department without human intervention. This is where the **100% Flexibility** of ACAS is configured and managed.

1. Performance Overview (Top Tiles)

- **Total Routed:** Tracks the overall volume of comments successfully moved through the system. This shows the scale of work ACAS is handling.



- **Active Rules:** Displays the number of custom logic paths currently running. Even with "0" manual rules, the core AI continues its work based on the **Avg Confidence (87%)** shown next to it.



- **Avg Confidence:** A high-level metric showing how reliably the AI is assigning tickets to their respective queues.



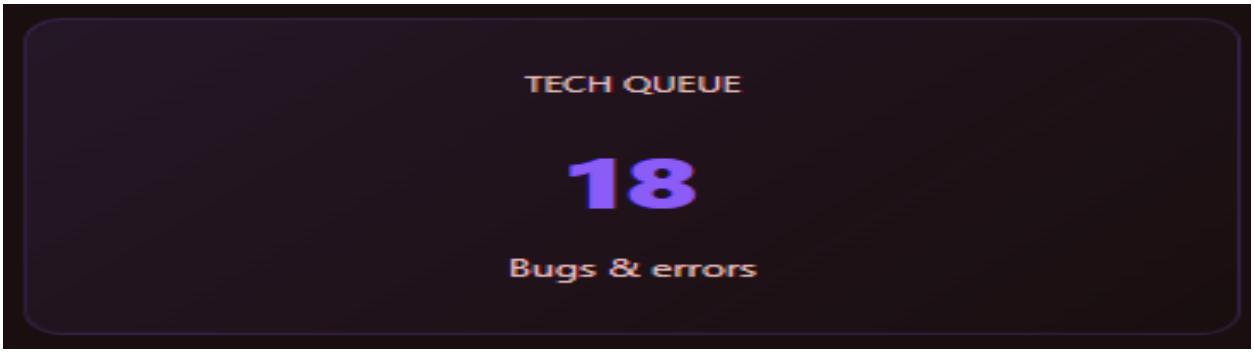
2. Routing Queues (The Destinations)

This section visualizes the departmental split. It shows how the "data avalanche" is broken down into manageable streams:

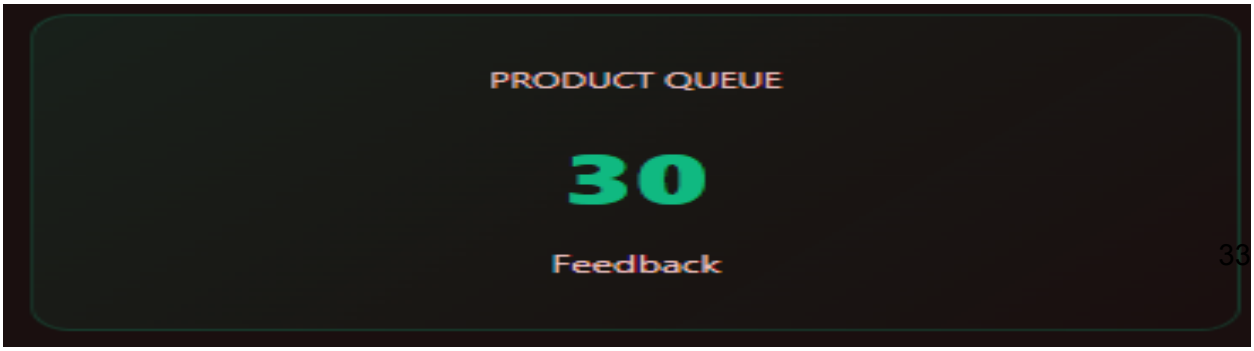
- **Support Queue (52 Help Requests):** Focuses on general assistance and customer maintenance.



- **Tech Queue (18 Bugs & Errors):** Captures technical failures, such as app crashes or website glitches.

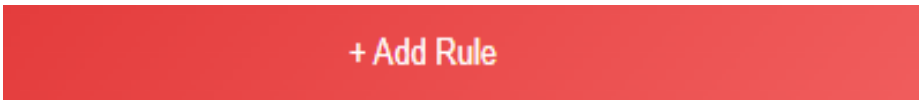


- **Product Queue (30 Feedback):** Gathers suggestions and feature requests to help the company grow.

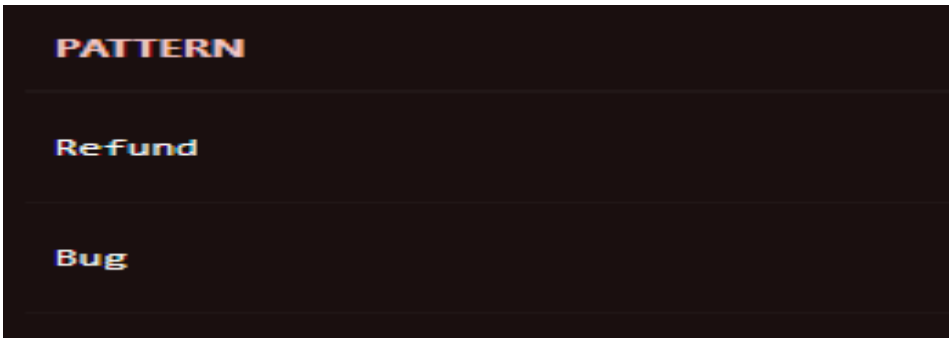


3. Rule Management (The "Control Logic")

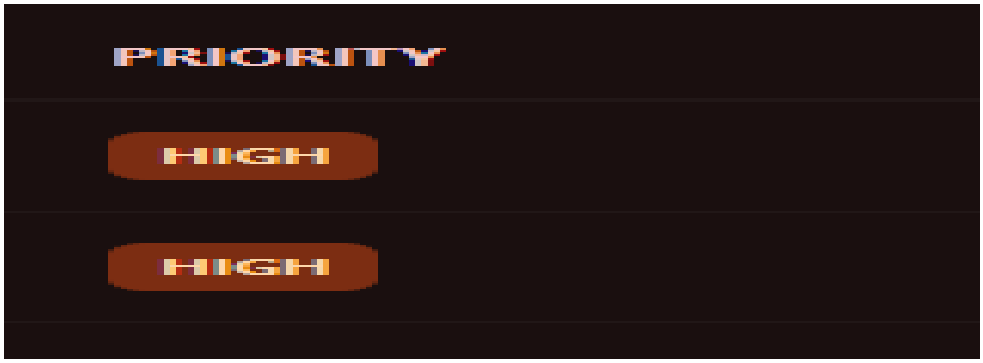
- **+ Add Rule Button:** A massive action bar that allows managers to create custom "If/Then" logic (e.g., *"If the comment contains 'Lost Luggage,' route to 'High Priority Support'"*).



- **Active Routing Rules Table:**
  - **Pattern:** Shows the specific keyword or intent the system is looking for (e.g., **Bug** or **Refund**).



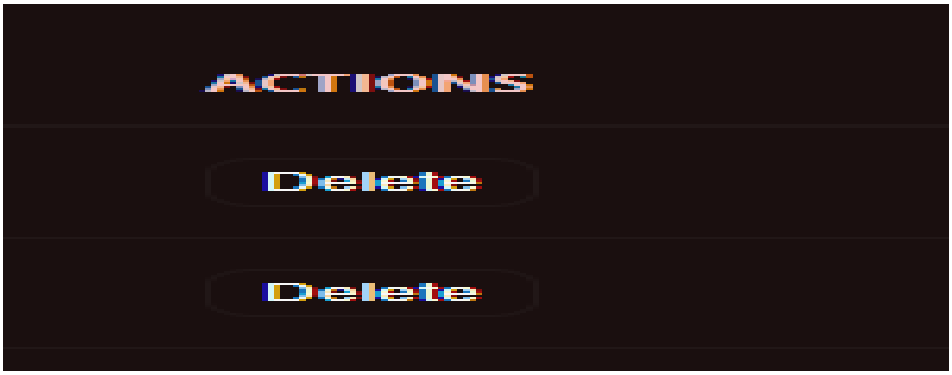
  - **Priority:** Sets the urgency level (e.g., **HIGH**) for that specific pattern.



- **Status:** A live indicator showing the rule is currently "Active" and protecting the department.

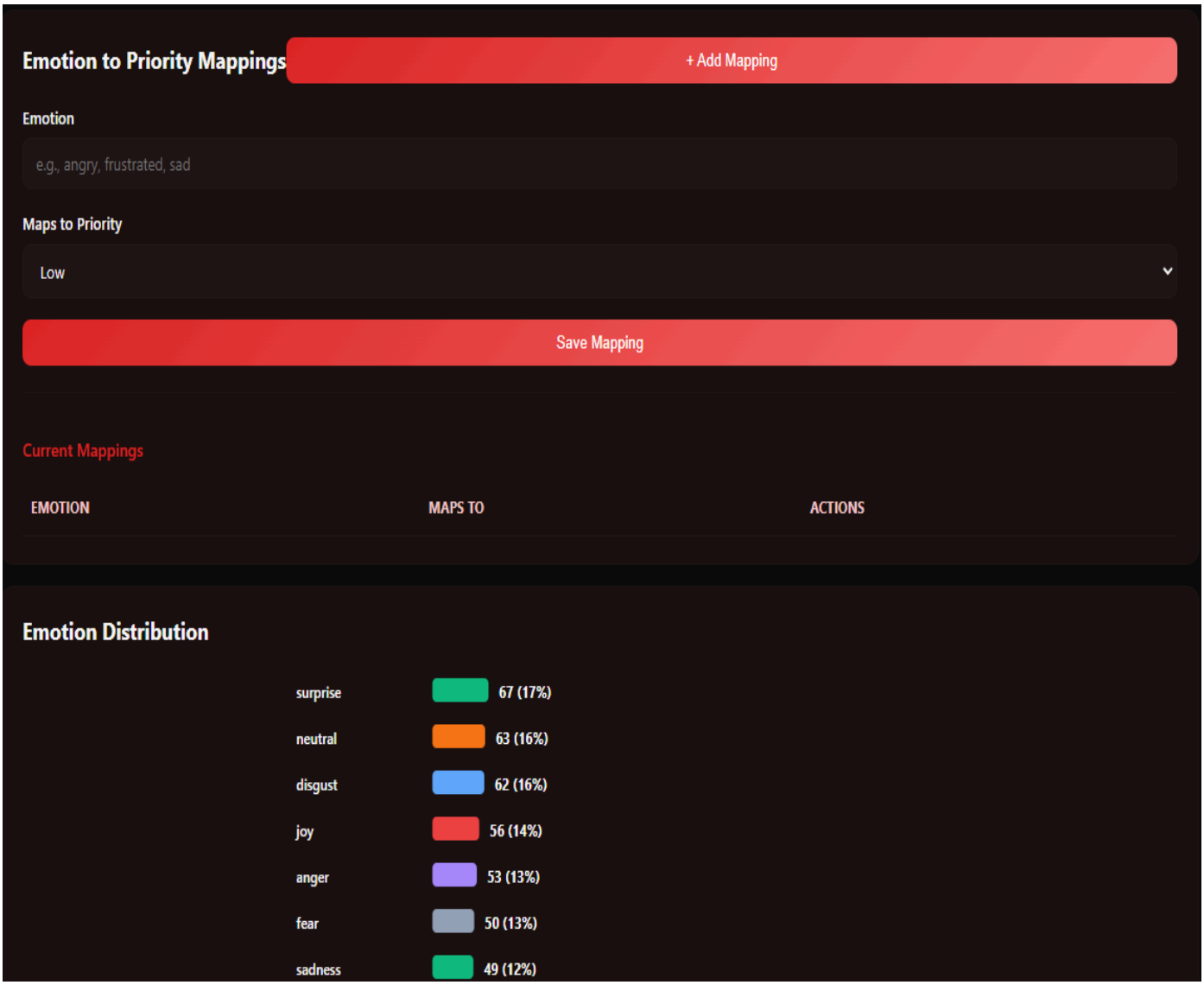


- **Actions:** Allows the manager to **Delete** or modify rules as the department's needs evolve.



Sentic Mapping Tab :

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The **Sentic Mapping Tab** is where the system's "Emotional Intelligence" is calibrated. This section allows the administrator to bridge the gap between human feelings and operational urgency, turning subjective emotions into objective business priorities

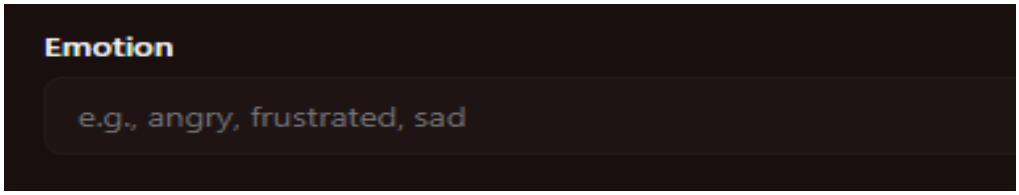
1. Emotion to Priority Mappings (The Decision Logic)

This is where the user defines how the AI should react to specific human moods.

- **+ Add Mapping:** An expandable tool to create new links between emotions and response levels.



- **Emotion Input Field:** Allows the user to specify an emotion (e.g., *angry, frustrated, sad*) that the system should watch for.



- **Maps to Priority Dropdown:** A critical setting where the user decides the urgency of that emotion. For example, "Anger" can be mapped to **High** priority, while "Neutral" stays **Low**.

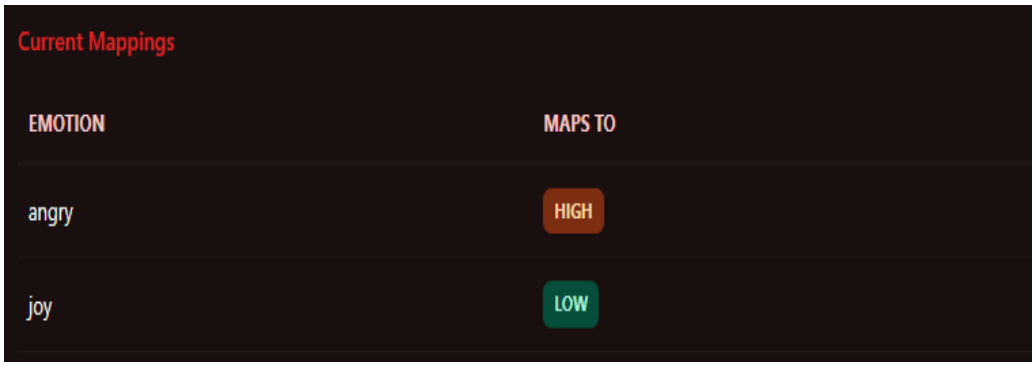


- **Save Mapping Button:** Commits the rule to the ACAS engine, instantly updating how future comments are triaged.

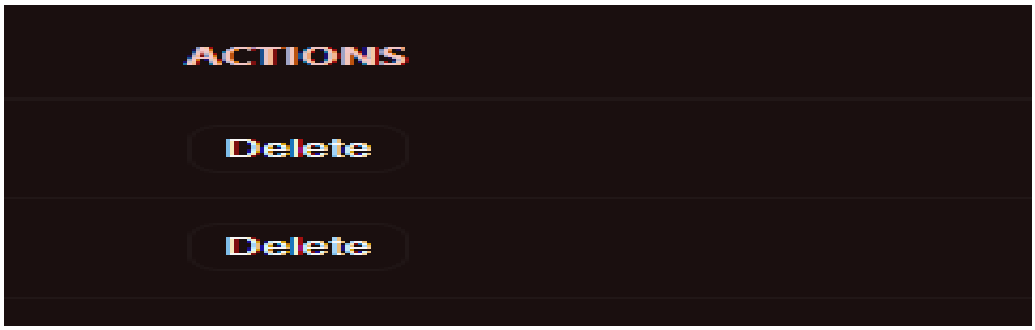


2. Current Mappings (Active Intelligence)

- This table lists all existing rules. It provides a transparent view of the system's "moral compass"—showing exactly which emotions trigger which priority levels.



- The **Actions** column allows for real-time adjustments as the organization's needs change (e.g., deciding that "Fear" should be handled more urgently during a crisis).



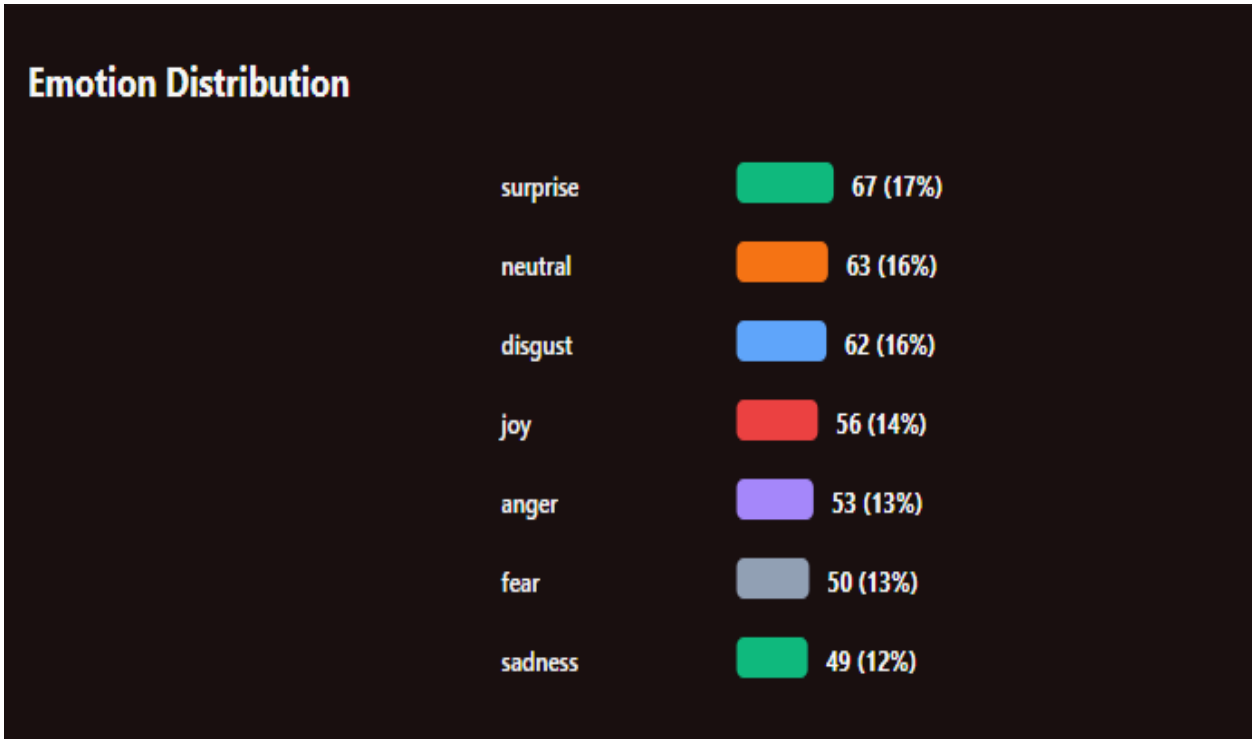
3. Emotion Distribution (Statistical Breakdown)

A detailed visual audit of the sentiment data processed by the system.

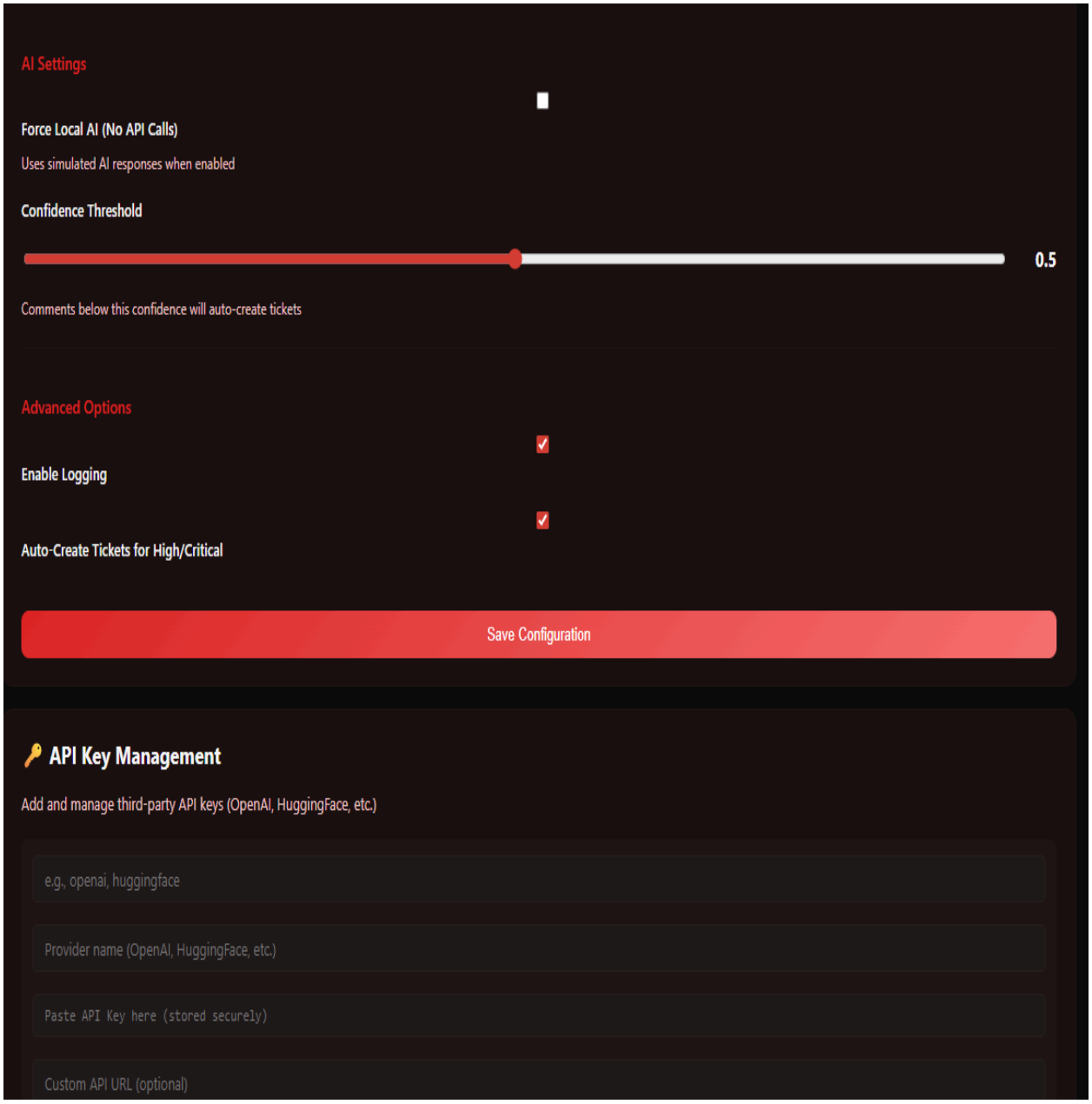
- **Granular Tracking:** Unlike simple positive/negative models, ACAS tracks seven distinct emotional states: **Surprise, Neutral, Disgust, Joy, Anger, Fear, and Sadness.**

- **Frequency & Percentage:** Displays both the raw count (e.g., 67 instances of Surprise) and the overall percentage (17%). This helps managers see the "Collective Mood" of their audience.

- **Trend Analysis:** By seeing that "Disgust" or "Anger" accounts for a high percentage (16% and 13% respectively), a manager can quickly identify that there is a systemic issue causing negative sentiment.



The Configuration Tab:



is where the administrative and technical parameters of ACAS are defined. This screen gives the user direct control over the AI's decision-making behavior and system security, ensuring the platform remains 100% flexible and reliable under different operational loads.

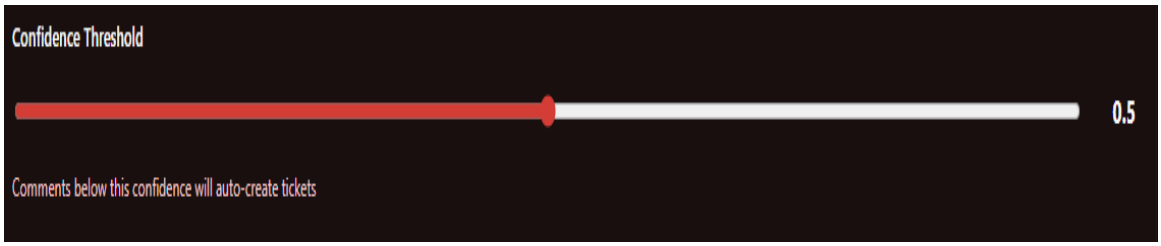
1. AI Core & Decision Logic

The top section of the configuration tab allows administrators to set the "behavioral rules" for how feedback is processed.

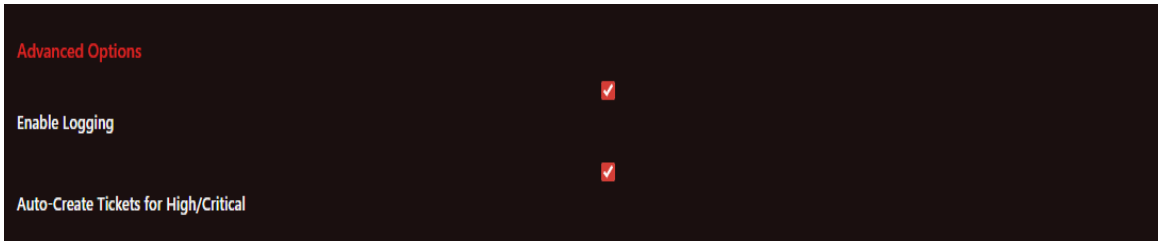
- **Force Local AI (Toggle):** A critical privacy feature. When enabled, all analysis stays within the local network (No API Calls), ensuring data sovereignty for sensitive departments like HR or Security.'



- **Confidence Threshold (Slider):** Currently set at **0.5**, this slider determines the AI's "autonomy." If the system's certainty in a comment falls below this mark, it triggers a manual review, preventing AI "hallucinations" or errors.



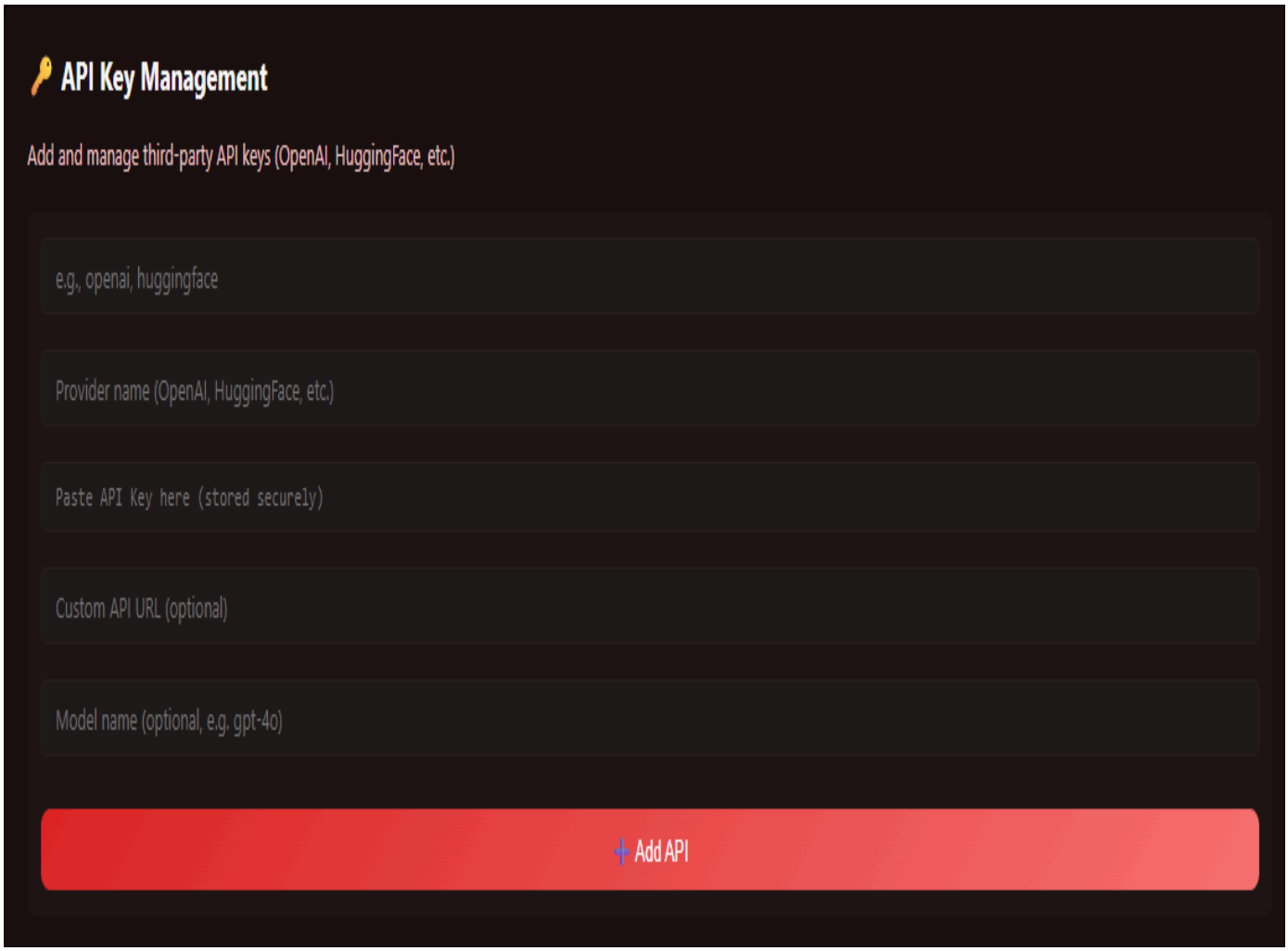
- **Advanced Automations:** * **Enable Logging:** Maintains a technical audit trail for system health monitoring.
    - **Auto-Create Tickets:** When enabled, any feedback tagged as **High** or **Critical** instantly generates a task, bypassing the main feed to ensure zero-delay intervention.



2. Global Intelligence & API Management

ACAS is designed as a **Modular Engine**, allowing it to connect to any industry-standard Large Language Model (LLM).

- **Third-Party Integration:** Secure input fields allow for the addition of API keys from providers like **OpenAI** or **HuggingFace**.
- **Customization Fields:** Administrators can specify custom API URLs or target specific models (e.g., **GPT-4o**), allowing the system to upgrade its intelligence without needing a full software rewrite.
- **+ Add API:** A streamlined action that validates and activates new intelligence streams across the platform.

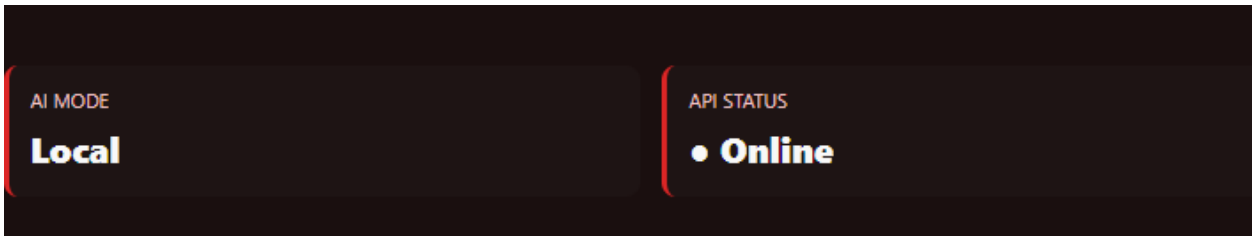


The footer provides a transparent look at the system's "Heartbeat," ensuring the technical foundation is solid.

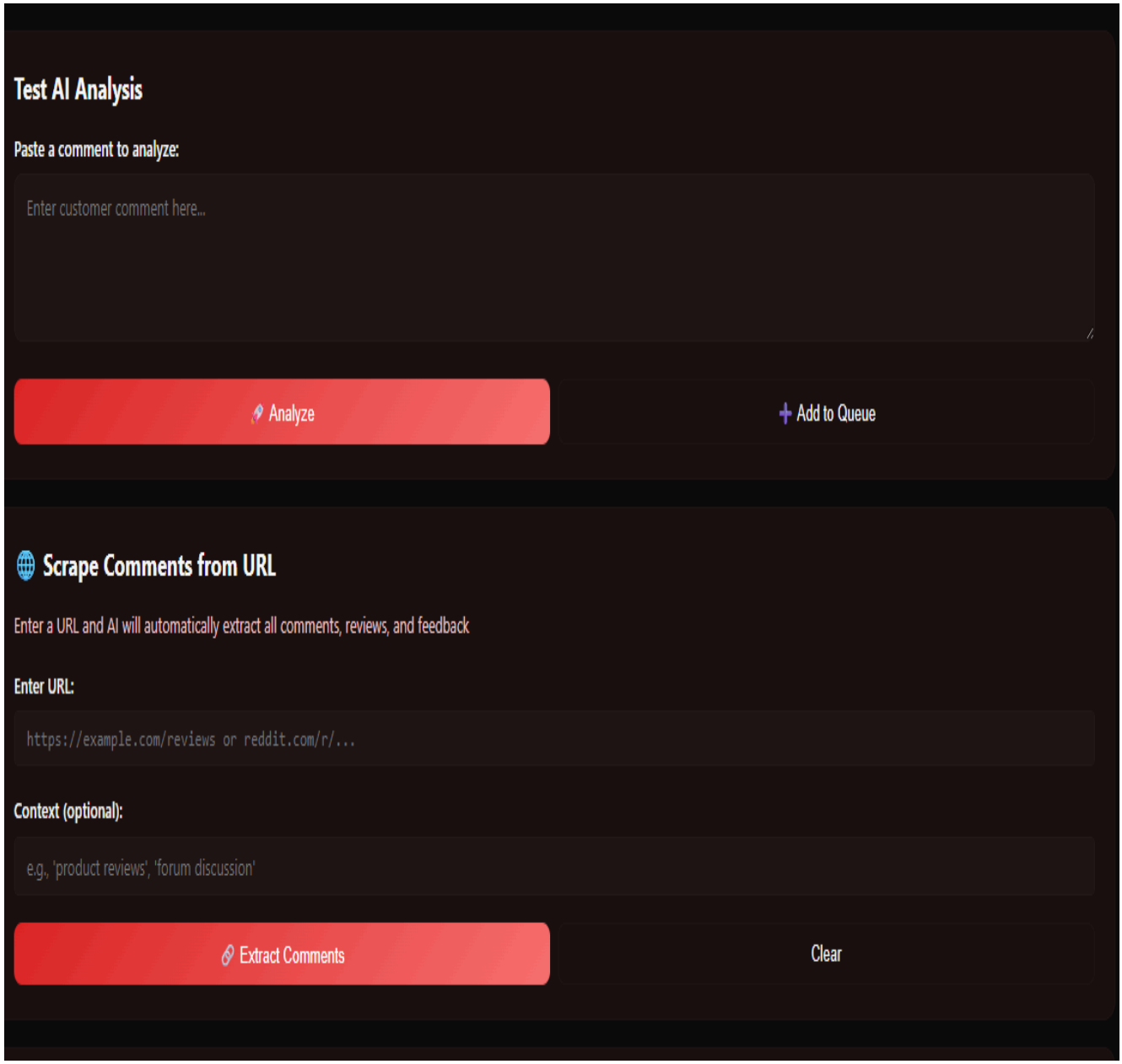
- **Database (SQLite):** Highlights a lightweight, high-performance local storage strategy, ideal for rapid data retrieval and portability.



- **AI Mode & API Status:** Real-time indicators showing the current processing state (**Local**) and the health of the connection (**Online**). This confirms that the "Digital Ear" is active and ready.



The Test Tab:

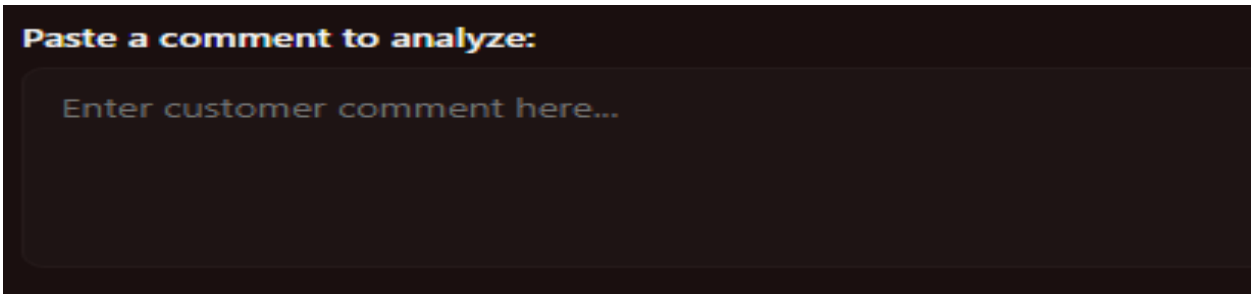


acts as a laboratory for developers and administrators to verify the system's accuracy before it hits live departments.

1. Test AI Analysis (Manual Validation)

This section allows for real-time "Sanity Checks" of the NLP engine.

- **Paste a comment to analyze:** A manual text input field where developers can enter specific, high-complexity phrases to test the AI's emotional detection.



- **Analyze Button:** Instantly processes the text and displays the predicted emotion, priority, and intent tags.



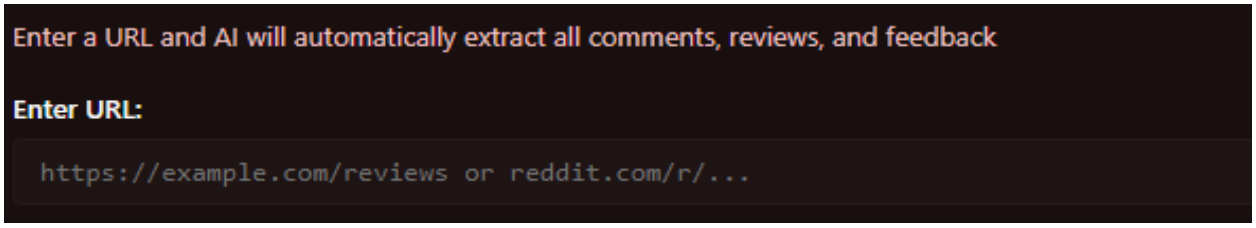
- **+ Add to Queue:** Once a test comment is analyzed, this button allows the user to push it into the real **Live Dashboard** feed to see how the system handles it in a live environment.



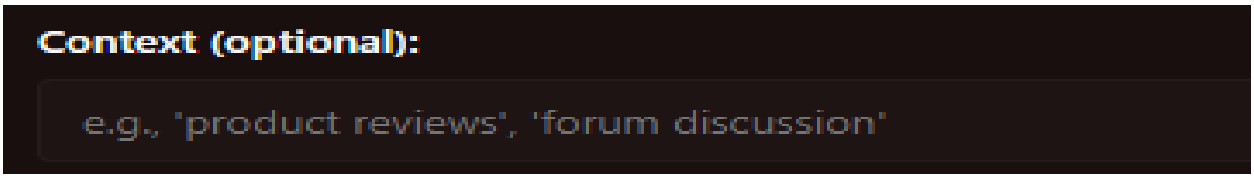
2. Scrape Comments from URL (External Data Ingestion)

This is a powerful expansion tool that proves ACAS can listen beyond internal company channels.

- **Enter URL:** Allows the system to target external websites, such as **Reddit communities, product review pages, or forum threads**.



- **Context (Optional):** A field to provide the AI with a "Perspective." For example, entering *'Customer reviews'* vs. *'Casual forum talk'* helps the AI adjust its sarcasm and slang detection models.



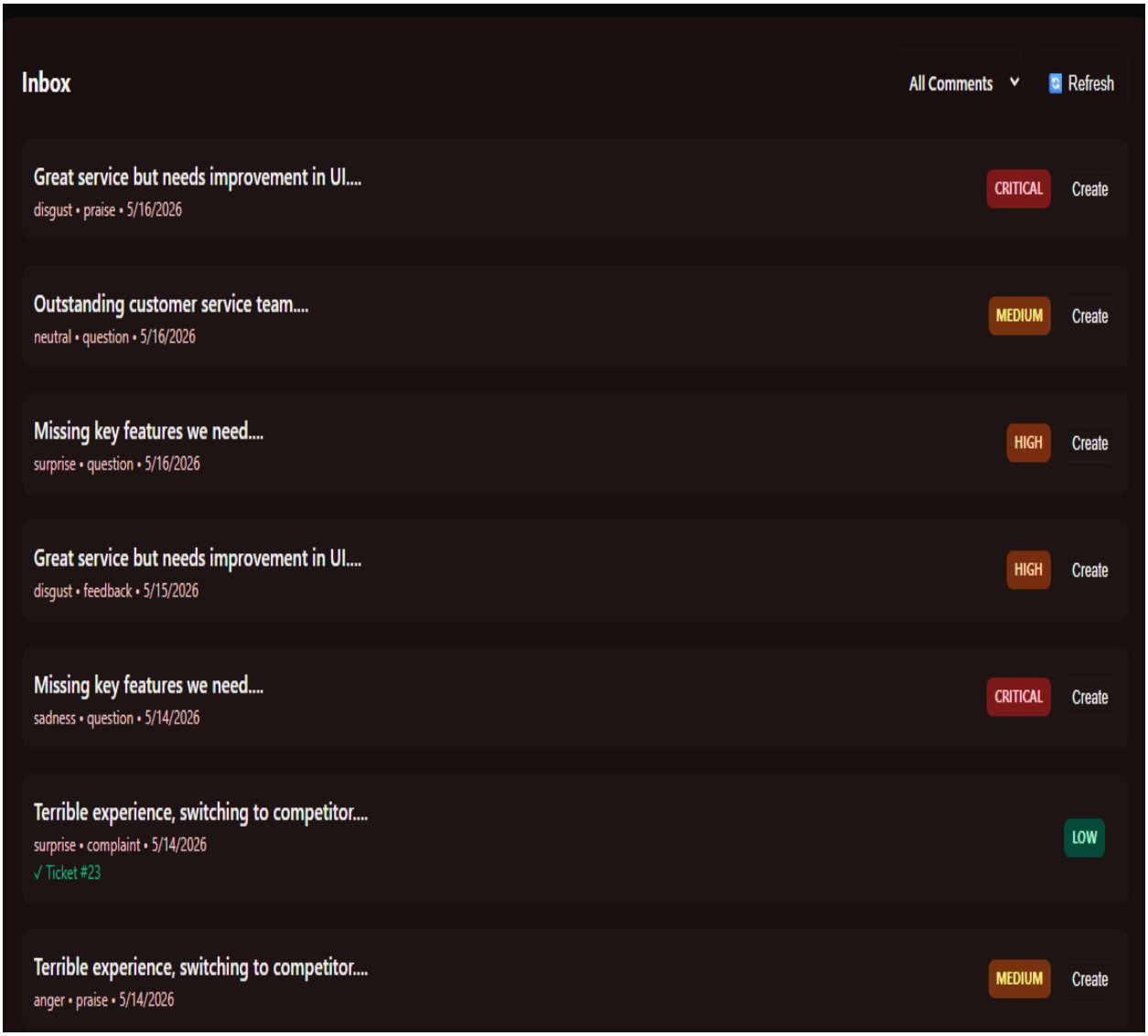
- **Extract Comments Button:** Triggers the web-scraping engine to pull raw text and feed it into the ACAS analysis pipeline.



- **Clear Button:** Instantly wipes the fields to prepare for a new data source.



The Inbox Tab:

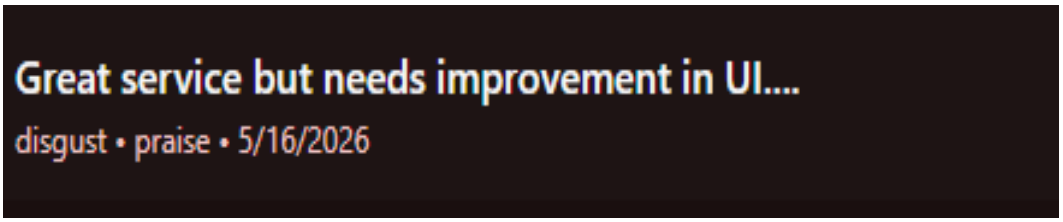


is the granular workspace where high-level AI analysis meets direct human action. This screen serves as the final destination in the operational loop, allowing staff to review specific feedback and convert analyzed comments into actionable tickets.

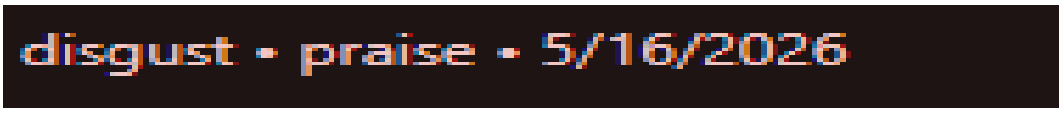
1. Threaded Feed (The "Conversation Stream")

Unlike raw data lists, the Inbox presents feedback as a clean, chronological feed, providing the full context for each interaction.

- **Comment Snippets:** Displays the core message of the feedback (e.g., *"Terrible experience, switching to competitor..."*).



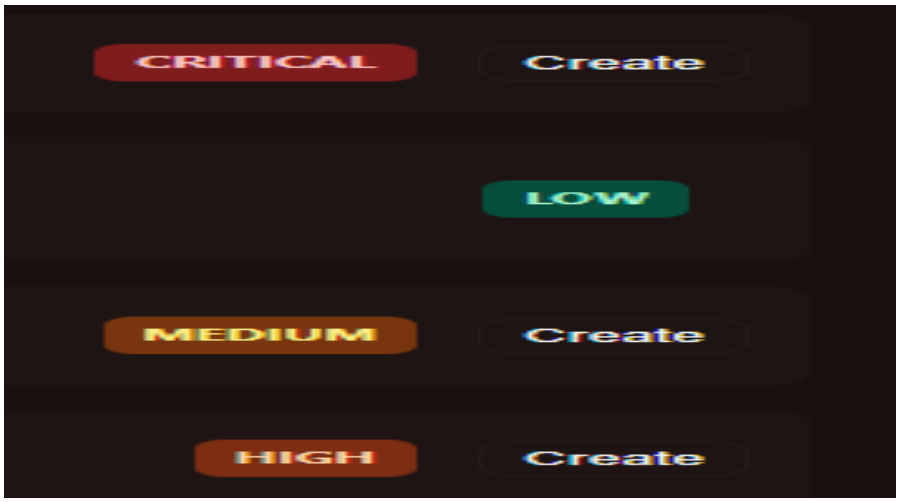
- **AI Metadata Tags:** Below each snippet, the system displays the AI's "Deep Analysis" tags:
  - **Emotion:** Identified feelings like **Disgust, Anger, Surprise,** or **Sadness**.
  - **Intent/Category:** Categorized as **Praise, Question, Complaint,** or **Feedback**.
  - **Date Stamp:** Tracks exactly when the feedback was received (e.g., **5/16/2026**) to ensure timely resolution.



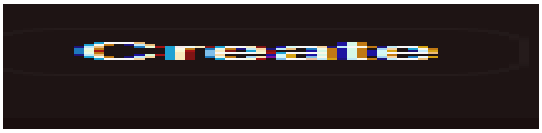
2. Priority & Action Controls

Every entry in the Inbox is equipped with immediate tools for workload management.

- **Priority Badges:** Color-coded status indicators—**CRITICAL (Red)**, **HIGH (Orange)**, **MEDIUM (Yellow)**, and **LOW (Green)**—allow staff to scan and prioritize their responses in seconds.

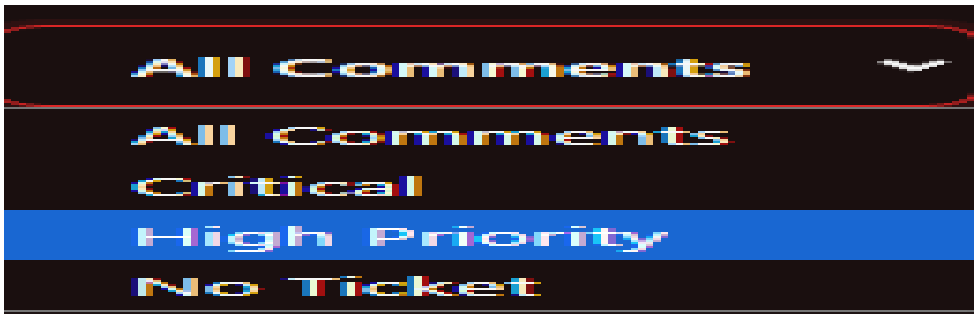


- **Create Ticket Button:** A primary action tool. If a comment requires follow-up, clicking "Create" formally enters the feedback into the organization's task management system.

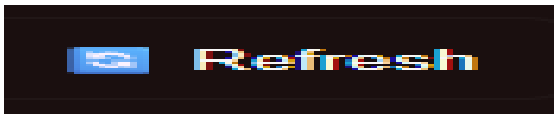


3. Global Navigation & Refresh

- **Comment Filter Dropdown (All Comments):** Allows the user to toggle between seeing all messages, only unread ones, or those already assigned to tickets.



- **Refresh Button:** A manual sync tool to pull the latest analyzed data from the ACAS engine into the local view.



The Technologies Used:

1. Development & Environment

- **Visual Studio Code (VS Code):** * **Where:** The primary Integrated Development Environment (IDE) for writing and debugging the full-stack application.
    - **Why:** It offers extensive support for Python and FastAPI, with a built-in terminal for running **PowerShell** commands and managing the virtual environment.
- **PowerShell:** * **Where:** Used for automation scripts, managing the **Python Virtual Environment (venv)**, and deploying the server locally.
    - **Why:** Provides a powerful interface for system-level operations and fast deployment during the testing phase.

2. Backend Architecture

- **Python:** * **Where:** The core programming language for the entire system logic.
    - **Why:** Python is the industry standard for AI and Data Science due to its extensive ecosystem of specialized libraries.
- **FastAPI:** * **Where:** The web framework used to build the API endpoints that connect the UI to the AI engine.
    - **Why:** It is one of the fastest Python frameworks available. Its **Asynchronous** nature allows ACAS to process large volumes of comments simultaneously without slowing down the user interface.
- **SQLite:** * **Where:** The database used for storing customer comments, ticket logs, and configuration settings.
    - **Why:** It is a lightweight, serverless database that stores everything in a single file.

3. Artificial Intelligence & Machine Learning

- **Hugging Face Transformers:** * **Where:** The core of the "Intelligence Layer" used to implement the Natural Language Processing (NLP) models.
    - **Why:** It provides access to state-of-the-art transformer models (like BERT or RoBERTa), which allow the system to understand the context and emotional nuances of human speech.
- **Multiclass Emotion Classification:** * **Where:** The custom-built model logic applied to every incoming comment.
    - **Why:** Unlike standard sentiment analysis, this allows the system to categorize feedback into specific buckets: **Anger, Joy, Frustration, Sarcasm, and Sadness.**
- **Supervised Learning:** * **Where:** The training phase of the AI model.
    - **Why:** By training the model on large, labeled datasets, the system reached a high level of accuracy (**87%+ confidence**), ensuring the auto-routing and priority tags are reliable.

4. Data Ingestion & UI

- **Web Scraping (BeautifulSoup & Requests):** * **Where:** Used in the **Test Tab** to pull data from external URLs like Reddit or review sites.
    - **Why:** It allows the system to analyze "huge amounts" of external public feedback, proving its flexibility beyond internal databases.
- **Data Visualization (Chart.js / Plotly):** * **Where:** Used in the **Dashboard** and **Sentic Mapping** tabs to render real-time emotional distribution and routing charts.
    - **Why:** It translates complex AI scores into simple, actionable visual data for managers.

Summary Table

Technology	Layer	Purpose
Python	Logic	Primary language for data processing and AI
FastAPI	Backend	High-speed API for real-time interaction
Transformers	AI	Context-aware emotion and intent detection
SQLite	Database	Efficient, portable storage of system data
VS Code	IDE	Central development and debugging hub
PowerShell	Automation Scripting	for deployment and server management
BeautifulSoup	Scraping	Extracting external feedback from public URLs.

Recommendations and Future Steps

1. Operational Recommendations

These suggestions focus on how an organization (like an airport or a large corporation) should best utilize ACAS today.

- **Human-in-the-Loop Validation:** While the AI maintains an **87% confidence rate**, it is recommended to keep a human moderator for "Critical" tickets. This ensures that the AI's emotional detection is periodically audited for accuracy.

- **Threshold Customization:** Organizations should use the **Configuration Tab** to adjust the Confidence Threshold based on their specific needs. High-security departments should set a higher threshold (e.g., **0.8**) to ensure only the most certain data is auto-processed.

- **Sentic Map Updates:** It is recommended to update **Emotion-to-Priority mappings** monthly. As public sentiment trends change, what was once a "Medium" priority issue (like a minor app bug) might become "High" priority during a major update.

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2. Technical Future Steps (The Roadmap)

These are the planned upgrades to make ACAS more powerful in future versions.

- **Multilingual Expansion:** * **Goal:** Integrate translation layers or multilingual Transformer models (like mBERT).
    - **Impact:** This would allow ACAS to analyze feedback in Arabic, French, or Spanish, which is vital for international environments like airports.
- **Voice-to-Text Integration:** * **Goal:** Add a module to process audio feedback from customer service hotlines or airport kiosks.
    - **Impact:** This expands the "Digital Ear" from written text to spoken word, covering all customer touchpoints.
- **Predictive Crisis Modeling:**
    - **Goal:** Implement a machine learning layer that identifies patterns *before* a crisis happens.
    - **Impact:** If the system notices a sudden 10% spike in "Confusion" and "Frustration" regarding a specific terminal, it can alert management to a potential operational failure before it escalates to "Anger."
- **Cloud Scalability:**
    - **Goal:** Move from a local **SQLite** database to a cloud-based solution like **PostgreSQL** or **MongoDB**.
    - **Impact:** This would allow ACAS to scale from handling thousands of comments to millions, supporting global enterprises with multiple branches.
- **Edge AI Deployment:**
    - **Goal:** Optimize the models to run on smaller, local hardware (Edge devices).
    - **Impact:** This would further enhance the **"Force Local AI"** feature, allowing the system to run on-site without any internet connection, maximizing data privacy and security.

Summary Table: Evolution of ACAS

Feature	Current State	Future Goal
Language	English+Arabic focus	Full Multilingual Support
Data Source	Text & Web Scraping	Voice, Video, and Image sentiment
Database	Local SQLite (Portable)	Distributed Cloud Database (Global)
Analysis	Real-time Triage	Predictive Analytics & Trend Forecasting

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Conclusion for Thesis: *"While ACAS currently provides a high-performance solution for real-time customer maintenance, its modular architecture allows for significant future growth. By integrating multilingual support and predictive modeling, ACAS will evolve from a reactive management tool into a proactive intelligence platform, capable of safeguarding an organization's reputation before issues even arise."*

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1. Core Frameworks & Technologies

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2. Machine Learning & Natural Language Processing

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  - Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., ... & Polosukhin, I. (2017). *Attention is all you need*. In Advances in neural information processing systems (pp. 5998-6008).
- **Hugging Face Library:**
  - Wolf, T., Debut, L., Sanh, V., Chaumond, J., Delangue, C., Moi, A., ... & Rush, A. M. (2020). *Transformers: State-of-the-Art Natural Language Processing*. Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing: System Demonstrations, 38–45.

- **Sentiment & Emotion Analysis:**
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